



Long sequence time-series forecasting with deep learning: A survey

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ARTICLE INFO

Keywords:

Time series forecasting
Long time series forecasting
Transformer
Data mining
Deep learning

ABSTRACT

The development of deep learning technology has brought great improvements to the field of time series forecasting. Short sequence time-series forecasting no longer satisfies the current research community, and long-term future prediction is becoming the hotspot, which is noted as long sequence time-series forecasting (LSTF). The LSTF has been widely studied in the extant literature, but few reviews of its research development are reported. In this article, we provide a comprehensive survey of LSTF studies with deep learning technology. We propose rigorous definitions of LSTF and summarize the evolution in terms of a proposed taxonomy based on network structure. Next, we discuss three key problems and corresponding solutions from long dependency modeling, computation cost, and evaluation metrics. In particular, we propose a Kruskal–Wallis test based evaluation method for evaluation metrics problems. We further synthesize the applications, datasets, and open-source codes of LSTF. Moreover, we conduct extensive case studies comparing the proposed Kruskal–Wallis test based evaluation method with existing metrics and the results demonstrate the effectiveness. Finally, we propose potential research directions in this rapidly growing field. All resources and codes are assembled and organized under a unified framework that is available online at https://github.com/Masterleia/TSF_LSTF_Compare.

1. Introduction

Time series forecasting (TSF) is a classical forecasting task that predicts the future trend changes of time series, and has been widely used in real-world applications such as energy [1], transportation [2], and meteorology [3]. Formerly, traditional statistical methods were effective ways to forecast time series. As early as 1927, Yule proposed the autoregressive [4] (AR) model for predicting univariate smooth time series. Since then, the moving average [5] (MA) model proposed by Walker and the autoregressive moving average (ARMA) [6] model, a combination of both, have been used to solve univariate smooth TSF problems. However, these traditional statistical methods often have many assumptions about time series data and are not effective in making valid predictions about reality, based on many prior assumptions such as stability, linear correlation, normal distribution, independence and other properties that limit the effectiveness in real-world applications. It is difficult to effectively capture the nonlinear relationships between time series. Traditional machine learning (ML) algorithms are effective solutions. Two types of classic algorithms, support vector machines (SVMs) [7] and adaptive boosting (AdaBoost) [8],

were developed in 1995 and have achieved great success in the field of TSF. They utilize feature engineering, which changes the organization of data with a sliding time window; i.e., data metrics, such as the minimum, maximum, mean, variance, etc., are calculated within the sliding window as new features for prediction. These algorithmic models solve the problem of predicting multivariate, heteroskedastic time series with nonlinear relationships to some extent. However, these models suffer from poor generalization, exhibiting limited predictive accuracy that can often only be improved with complex artificial feature engineering. The rapid development of deep learning (DL) in recent years as shown in Fig. 1, it has greatly improved the nonlinear modeling capabilities of TSF method. DL is called an effective solution for TSF, and it is widely used in finance, energy, meteorology, transportation and medical fields. It has also extended many other problems related to TSF, such as hierarchical time series forecasting [9], intermittent time series forecasting [10], sparse multivariate time series forecasting [11] and asynchronous time series forecasting [12,13], and has even extended some multiobjective and multigranular forecasting scenarios [14] and

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<https://doi.org/10.1016/j.inffus.2023.101819>

Received 3 December 2022; Received in revised form 19 April 2023; Accepted 21 April 2023

Available online 28 April 2023

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Table 1
Recent survey comparison.

Survey	Theme	Contribution
Bianchi F M [20]	Focus on RNN in short-term load forecasting	- A survey on short-term load forecasting. - The case study about reviewed models.
Manibardo E L [21]	Focus on DL in road traffic forecasting	- Critically analyzing the state of the art in what refers to the use of DL for Intelligent Transportation Systems research. - An extensive experimentation comprising.
Lara-Benítez P [22]	Focus on DL in TSF	- A exhaustive review about DL in TSF. - An open-source deep learning framework for TSF and a comparative analysis.
Deb C [23]	Focus on energy consumption forecasting based on ML	- A comprehensive review and comparison of ML in energy consumption forecasting. - Provides constructive future direction.
K Benidis [24]	The main focus is to educate, review and popularize the latest developments in NN-driven forecasting	- The breadth and depth survey on DL for TSF. - Digging deeper and find out future directions.
Our	Focus on DL in long sequence time-series forecasting	- Deep explanations from multiple perspectives about LSTF definition. - New taxonomy and comprehensive review for LSTF. - New performance evaluation for LSTF. - Abundant resources of TSF and particularly in LSTF, such as datasets, metrics, open source library, eg. - Future directions for LSTF.

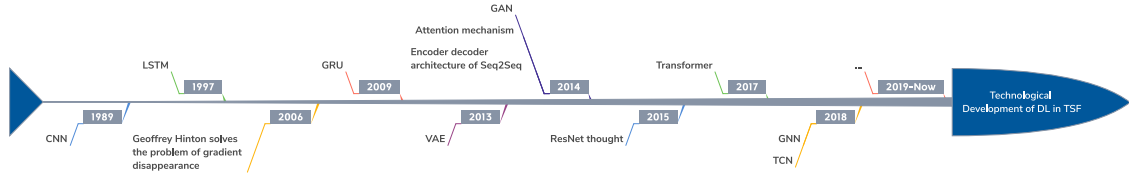


Fig. 1. Technological Development of DL in TSF.

multimodal time series forecasting scenarios [15,16]. With the development of technology, a task gradually opens to using more historical data to predict the longer-term future, that is long sequence time-series forecasting (LSTF) [17–19], and researchers have also explored commonly feasible solutions to the LSTF.

LSTF has been in greater demand for applications in financial, energy, meteorological, transportation, and medical scenarios, such as electricity usage planning [25] and financial long-term strategic guidance [26]. The LSTF is a critical component in assisting individuals to better plan for the future by forecasting outcomes further in advance. Nevertheless, prior algorithms have demonstrated unsatisfactory performance when attempting to forecast longer-term predictions with an increased number of input sequences. Specifically, such algorithms are prone to a significant decline in inference performance and predictive accuracy due to the growing sequence length. Furthermore, they struggle to efficiently extract more effective dependencies from the data, posing an urgent problem in time series forecasting. However, with the remarkable advancement in computational resources and deep learning techniques, this challenge has been gradually and effectively addressed.

Many researchers have studied LSTF from the perspectives of traditional statistics and machine learning [27–31]. With the development of DL, especially the emergence of Transformers [32], the ability to deal with long-sequence problems has attracted wide attention in the industry. Researchers started to study the applications of Transformers in the field of TSF, and works such as [17,33] demonstrated the effectiveness of Transformer-based models for LSTF. The recent work of [34] summarized the applications of Transformer in TSF. The parallel computing approach in the Transformer-based architecture speeds up the inference in LSTF and the capacity of long series data that the model can handle, while the number of operations required for self-attention to compute the association between two time points does not grow with the distance between the two time points making it easier

to capture long-term dependencies between the data. However, the expensive training and deployment costs of these models make them unaffordable when applied to real LSTF problems. Transformer-based models with large parameters are also prone to overfitting in some data distributions. Thus, there are also many other nontransformer-type works [29,35–41] that are studying TSF in general and LSTF problems in particular. As the LSTF is receiving more and more attention, its related research is increasing. In particular, in view of the current status of the existing work, we found the following three problems in the field of LSTF, which have not been standardized and sorted out in other works

- **Problem 1: To our knowledge, long sequence time-series forecasting still lacks a comprehensive overview in data mining and deep learning fields.** In particular, the rapid development of LSTF based on deep learning technology has created a pressing need for an overview of methods, data, and metrics to inspire researchers who are interested in entering this rapidly developing field, as well as experts who wish to compare LSTF methods. Many existing studies survey time series forecasting [20–24], these reviews tend to categorize and summarize time series forecasting research under one overarching theme, as shown in Table 1, without distinguishing between LSTF and other forecasting approaches.
- **Problem 2: Long sequence time-series forecasting lacks deep explanations from multiple perspectives.** Generally, LSTF is defined as forecasting the distant future [17]. However, such a definition is macroscopic, and lacks a comprehensive and nuanced explanation of LSTF in the existing literature. We are unsure about the specific forecasting task that qualifies as LSTF.
- **Problem 3: Long sequence time-series forecasting still lacks technical analysis and performance comparison of different neural networks under a unified framework.** The traditional

MSE and MAE metrics only reflect the error between the true and predicted values. However, in LSTF, with the increase of predicted horizon, it is not reliable to compare the accuracy of models purely in terms of the numerical magnitude of the evaluated metrics. For example, we cannot compare the superior results [42] of PatchTST/42 and DLinear for MSE=0.202 and MSE=0.203. We need to prove the superiority of one model over the other in a statistical sense. However, there is no relevant performance evaluation.

In order to address the above challenges, we have provided a comprehensive overview of LSTF, and this article is applicable both to interested researchers who want to enter this fast growing field and to experts who want to compare LSTF models. To cover a broader range of approaches, we consider LSTF as a task for forecasting more distant futures in TSF. In other words, we consider as the LSTF literature in which the contribution, motivation, and the problem addressed is to predict a more distant future in TSF. Thus, LSTF is part of TSF, and the difference between spatio-temporal forecasting and LSTF is that they view the problem from different perspectives and they have an intersection part. We retrieved data in a predefined manner from the Web of Science (WoS) Core Collection database and DataBase systems and Logic Programming (DBLP). The document types were confined to article or review. After filtering the articles by reading their abstracts, we also selected closely related references to complement them.

The contributions of this paper are summarized as follows:

- (1) **Deeply multi-Perspective LSTF definition explanations:** The existing definition of LSTF is quite macroscopic and ambiguous. Clearly defining the LSTF is challenging. To address it, we provide deep explanations from multiple perspectives: relative and absolute concepts.
- (2) **New Taxonomy and Comprehensive Review:** We propose a new taxonomy of LSTF research. We classify them into RNN-based model, CNN-based model, GNN-based model, Transformer-based model, Compound model and Miscellaneous methods according to their main predictive network structure characteristics. In addition, we summarize several key problems of LSTF and investigate the related work and effective solutions of each key problem.
- (3) **New Performance Evaluation:** To enhance fair and effective performance evaluation of LSTF, we apply Kruskal–Wallis test and propose a new LSTF prediction performance evaluation method from two perspectives.
- (4) **Abundant Resources:** We have collected abundant resources of TSF and LSTF in particular. We have classified five application domains and collected relevant datasets for each domain, giving the relevant open source links for each dataset. We also collected relevant evaluation metrics, classified them into scale-dependent, scale-independent, and scaled errors, and discussed the properties of each metric. Finally, we selected SOTA (Pyraformer, FEDformer, Autoformer, Informer, Reformer, Transformer, MTGNN, Graph WaveNet, LSTNet) models in recent years and based on our research. Based on the datasets and evaluation metrics of our research, we give a technical analysis and performance comparison of different neural networks under a unified framework in LSTF, and an open source library.
- (5) **Future Directions:** Based on the results of our research and experiments, we have summarized four possible future research directions.

The motivation and line logic of this paper are shown in Fig. 2. In this paper, definitions of time series data, TSF tasks, and deeply multi-angle definition explanations for LSTF are given in Section 2. In Section 3, we review the relevant research and the evolution of DL networks in the LSTF field in recent years and classify the related studies according to the technical contributions of each work. Section 4

Table 2

A summary of notations.

Notations	Descriptions
X	Denotes a multidimensional time series matrix
$\vec{x}$	Denotes a time series vector
$f(\cdot)$	Denotes a time series forecasting model
$\hat{X}_{i,t+1}$	Denotes the predicted value of the i th feature at $t+1$ time step
$X_{i,t-n:t}$	Denotes the historical value of the i th feature from time step $t-n$ to t
t	The current time step
l	Indicates the interval between time series data
m	Indicates sample size
$\hat{y}$	Indicates the predicted value
y	Indicates true value
$\bar{y}_i$	Denotes the mean of the real sample data
y_{max}	Denotes the maximum value of the real sample data
y_{min}	Denotes the minimum value of the real sample data
$mean(\cdot)$	Denotes the mean value
H	Denotes the statistic in the K-W test
R_i	denotes the sum of the rank of n_i observations $\{X_1, \dots, X_{n_i}\}$ of the i th sample in this arrangement
H'	Denotes the corrected K-W test statistic
α	Indicates the significance level
V	Degree of freedom
χ^2	Denotes the cardinal distribution
$\chi^2_{\alpha,v}$	Denotes the upper lateral quantile of the chi-square distribution

summarizes the key problems encountered in LSTF, sorts out the research directions and effective solutions for each key problem in the current works, and proposes our new LSTF evaluation approach for performance evaluation problems. Section 5 summarizes the relevant open source datasets in various domains of the five application areas. Section 6 summarizes and discusses the predictive performance evaluation metrics and classifies them. In Section 7, we conduct a case study to compare the SOTA models that have been developed in recent years to evaluate and compare their LSTF performance. In Section 8, we discuss future research directions.

2. Problem definition

In this section, we present the definitions of time series data and their characteristics. TSF tasks and their classification methods are given based on their properties. We also note that the terms “variable” and “feature” are used interchangeably in this paper. Unless otherwise noted, the notations used in this study are shown in Table 2.

2.1. Time series data

Time series data exist in various fields in real life, and it is particularly important to master the distribution characteristics of time series data and predict their future laws. By predicting future gold price fluctuations or future stock market price fluctuations to obtain large benefits and by predicting future weather or predicting future traffic flows to benefit people's lives, it can be seen that grasping the future development law of time series data is very meaningful. So what are time series data?

Time series data are series of finite or infinite random variables (sequence data) arranged on the basis of time. This type of data reflects the changing state of a certain thing or phenomenon over time. In addition to digital data in the traditional sense, if sequence data such as video data, audio data, and voice data are arranged according to time, they can be regarded as time series data. That is, any data observed and recorded in a time sequence can be regarded as time series data. An N -dimensional time series matrix $\hat{X}_{1:N,t-n:t}$ with n samples can be expressed formally as follows:

$$\hat{X}_{1:N,t-n:t} = \{\vec{x}_{1,t-n:t}, \vec{x}_{2,t-n:t}, \dots, \vec{x}_{N,t-n:t}\} \quad (1)$$

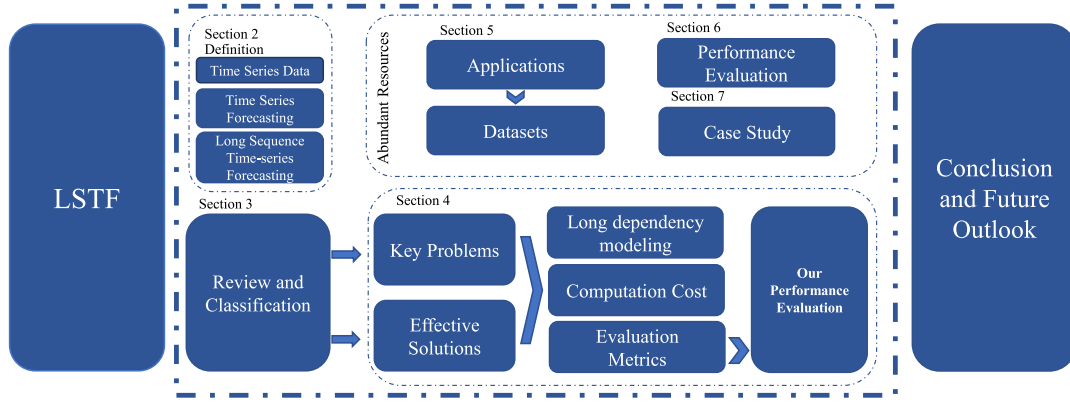


Fig. 2. Article Structure.

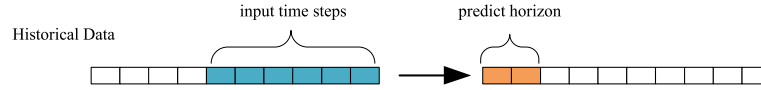


Fig. 3. Univariate Forecasting.

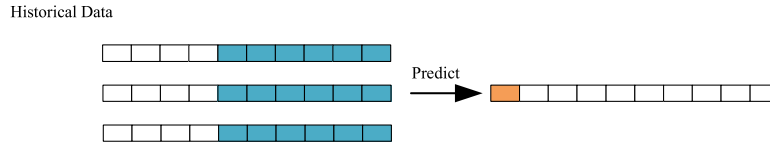


Fig. 4. Covariate Forecasting.

Among them, $\tilde{x}_{1,t-n:t}$ is represented as a vector, $\tilde{x}_{1,t-n:t} = \{x_{1,t-n}, x_{1,t-n+1}, \dots, x_{1,t}\}$, and $x_{1,t-n}$ is denoted as a time series point at time $t-n$ of a time series. Thus, $\tilde{X}_{1:N,t-n:t}$ denotes time series data consisting of multiple time series.

Time series data exist in major fields such as finance, energy, meteorology, transportation, and medical care. They all have the following common characteristics.

- **Trend (T):** A trend refers to the change in a time series with the progress of time or the change in an independent variable, and data may present a relatively slow long-term change trend of continuous rising or falling.
- **Periodicity (C):** Periodicity means that the given time series and its independent variables changes with time, and its data show a regular rise or fall.
- **Seasonality (S):** Seasonality means that the time series changes with natural seasons and time periods, and its data exhibit regular rises or falls with season changes. The definition of seasonality is very similar to that of periodicity, and many people confuse the two. However, in fact, there is a significant difference between them. Seasonality represents that the time series changes at a known and fixed frequency with the influence of seasons, while periodicity means that the time series rises or falls with an unfixed frequency.
- **Randomness (I):** Randomness is manifested in the fact that time series data points show irregular changes, but the whole series satisfies statistical laws; these changes are sometimes called residuals.

In summary, time series have four characteristics: trend, periodicity, seasonality and randomness. In addition, time series can also be classified according to other characteristics of TSF tasks, such as data distribution characteristics. There are also stationary time series and nonstationary time series in time series data. Stationarity means that the mean and variance of the time series are consistent at different

times. In many cases, methods such as time series decomposition are used before forecasting to convert nonstationary time series into stationary time series, which can improve the accuracy of forecasting. Notably, according to the nature of time series, they can also be divided into hierarchical time series [9], fuzzy time series [43] and so on. However, this paper only discusses LSTF under precise time series. In the next section, we introduce the definition of TSF tasks.

2.2. Time series forecasting

Many analysis and processing tasks are available for time series data, such as time series classification and TSF. This paper focuses on the TSF task. The TSF task is a classic problem that involves using one or more time-ordered features of something in a prior period of time to predict the features of that thing in a future period of time. This task is different from the general regression analysis prediction model. A TSF model needs to capture the sequence of the series data to grasp the future trend of the sequence data. The same sequence data are changed in order, and the result generated by inputting the time series data into the time series model is completely different. If there is an i th group time series with n sample data $X_{i,t-n:t} = \{x_{i,t-n}, x_{i,t-n+1}, \dots, x_{i,t}\}$, We take the historical data of this time series to predict its future value at $t+1$ time step, then the result $\hat{X}_{i,t+1}$ is:

$$\hat{X}_{i,t+1} = f(X_{i,t-n:t}). \quad (2)$$

where $f(x)$ represents the forecasting model, $x_{i,t} \in \mathbb{R}$.

However, in different TSF tasks, due to the different numbers of required input features, the different numbers of features predicted by outputs, the different lengths of output time series, and the different distributions and data characteristics of time series data for each feature [44,45], there are a wide variety of TSF tasks. The classification methods employed for time series tasks are given below according to the characteristics of different tasks.

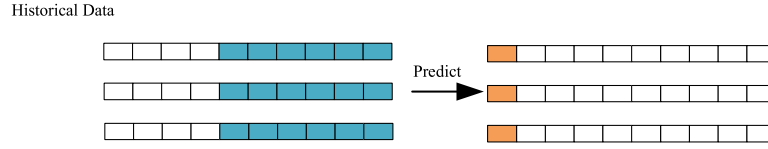


Fig. 5. Multivariate Forecasting.

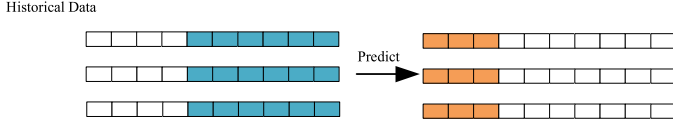


Fig. 6. Multi-Step Forecasting.

(1) Univariate/Covariate/Multivariate Forecasting

According to the different number of features required for the time series forecasting task and the different number of features of the forecasting target, we can classify the forecasting tasks into univariate forecasting tasks, covariate forecasting tasks and multivariate forecasting tasks.

• Univariate:

We refer to the univariate forecasting task in which both the number of input features and the number of predicted features required are univariate, as shown in Fig. 3, in which the future value of a single variable is predicted by inputting historical data for that variable, and the size of the predicted future window is referred to as the predict horizon in this paper. The mathematical expression of the univariate forecasting task is as follows.

$$\hat{X}_{t+1} = f(X_{t-n:t}) \quad (3)$$

Where t is the current time step, n denotes the input time steps size, the task is expressed as using the historical data of the previous n time steps at moment t to predict the feature values at $t+1$ time step in the future.

• Covariate:

Similarly, when predicting the feature of a single variable, it is sometimes necessary to model the association of features among multiple variables to predict the future of a single variable, and we refer to these variables as covariates, and such a forecasting task is called a covariate forecasting task. Note that the number of features predicted in the covariate forecasting task is still single as in univariate prediction, but it has multiple input features, as shown in Fig. 4. In the case of single-step forecasting, the mathematical expression of the covariate forecasting task is shown below.

$$\hat{X}_{t+1} = f(X_{i,t-n:t}) \quad (4)$$

where i is the number of input features, n denotes the input time steps size, t denotes the current time step, $i > 1$.

• Multivariate:

In addition to the above two cases, there exist forecasting tasks that model the feature associations among multiple variables in order to predict the feature values of multiple variables, which we refer to as multivariate forecasting tasks. Unlike the above two types of tasks, the inputs and outputs of this task are multivariate, as shown in Fig. 5. The mathematical expression of the multivariate forecasting task within single-step forecasting is shown below.

$$\hat{X}_{j,t+1} = f(X_{i,t-n:t}) \quad (5)$$

Where i is the number of input features, n denotes the input time steps size, j is the number of output features, $j \leq i$, and t is the current time step.

(2) Single-Step Forecasting/Multi-Step Forecasting

We refer to the task that predicts a single time step in the future as a single-step forecasting task, and the task that predicts multiple time steps in the future as a multi-step forecasting task, which is shown in Fig. 6. The mathematical expressions of the single-step forecasting task and the multi-step forecasting task in the context of multivariate forecasting are as follows.

• Single-Step Forecasting:

$$\hat{X}_{j,t+1} = f(X_{i,t-n:t}) \quad (6)$$

• Multi-Step Forecasting:

$$\hat{X}_{j,t+1:t+h+1} = f(X_{i,t-n:t}) \quad (7)$$

Where i is the number of input features, j is the number of output features, $j \leq i$, n denotes the size of input time steps, t is the current time step, and h denotes the predicted horizons.

Recursive forecasting is a way to achieve multi-step forecasting, which is mainly through each single-step forecasting to get the prediction value at time $t+1$, and then input the prediction value at time $t+1$ and the previous historical sequence into the prediction model to get the prediction value at time $t+2$, and through this iteration to get the prediction value at time $t+n$ to achieve the effect of multi-step forecasting, as shown in Fig. 7. The mathematical definition of recursive forecasting is as follows.

$$\begin{aligned} \hat{X}_{i,t+1} &= f(X_{i,t-n:t}) \\ \hat{X}_{i,t+2} &= f(X_{i,t-n+1:t+1}) \\ &\vdots \\ \hat{X}_{i,t+h+1} &= f(X_{i,t+h-n:t+h}) \end{aligned} \quad (8)$$

Where i is the number of input features, n denotes the size of input time steps, t is the current time step, and h denotes the predicted horizons. However, this method often leads to the accumulation of prediction errors, resulting in low prediction accuracy for the final model.

(3) Isometric/Nonisometric Interval Forecasting

In addition to the above classification methods, the developed approaches can also be divided according to whether the prediction step size of the forecasting task is equidistant. We refer to tasks with equal prediction steps as isometric interval forecasting tasks, and tasks with unequal prediction steps as nonisometric interval forecasting tasks. In a case with a univariate single-step forecasting, the mathematical expressions of isometric interval forecasting tasks and nonisometric interval forecasting tasks are as follows:

• Isometric Forecasting:

$$\hat{X}_{t+1} = f(x_{i,t-n}, x_{i,t-n+1}, x_{i,t-n+2}, \dots, x_{i,t}) = f(X_{i,t-n:t}) \quad (9)$$

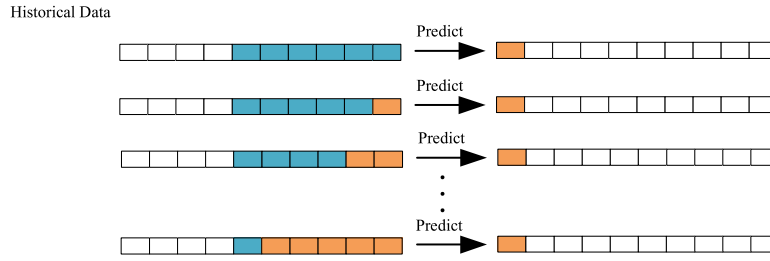


Fig. 7. Recursive Forecasting.

- Nonisometric Forecasting:

$$\hat{X}_{t+1} = f(x_{i,t-n}, x_{i,t-n+l_1}, x_{i,t-n+l_2}, \dots, x_{i,t}) = f(X_{t-n:t}) \quad (10)$$

Where t is the current time step, and $\{l \leq n | l_1, l_2, l_3, \dots, l_n \in \mathbb{N}^+\}$. There are few nonisometric forecasting cases, so we only discuss isometric forecasting.

(4) Homogeneous/Heterogeneous Forecasting

All the above classification methods are based on homogeneous time series, while a large number of heterogeneous time series exist in the fields of unmanned, smart home, robotic servants and aviation. For example, Li et al. [45] combined synchronous time series and asynchronous event sequences for prediction. Thus, the forecasting task can also be divided into homogeneous time series forecasting tasks and heterogeneous time series forecasting tasks. Heterogeneous time series contain different types of embedded, statistical patterns [44], such as the time series consisting of continuous and categorical variables, we refer to the task of combining different heterogeneous data for forecasting as heterogeneous time series forecasting. The mathematical expression of heterogeneous time series forecasting in the context of single-step forecasting is as follows.

$$\hat{X}_{t+1} = f(X_{t-n:t}, S_{i,\tau}) \quad (11)$$

Where $S_{i,\tau}$ is the i th heterogeneous time series of the input, $S_{i,\tau} = \{s_{i,\tau_k} | \tau_k < t\}$, and t is the current time step. In this paper, only the case of homogeneous time series prediction is discussed

With reference to the above classification, a time series forecasting task can be defined as univariate single-step forecasting/multi-step forecasting, covariate single-step forecasting/multi-step forecasting and multivariate single-step forecasting/multi-step forecasting according to its characteristics. In this paper, the default is all the cases of isometric homogeneous time forecasting.

2.3. Long sequence time-series forecasting

The definitions of time series and TSF tasks are introduced above, and this section discusses the definition of LSTF based on investigating the literature and industry standards in various fields.

For the definition of the LSTF, we have reviewed studies of LSTF in recent years and its applications in various areas; and it was usually defined as forecasting a more distant future [17,46]. However, this definition is quite macroscopic and ambiguous. Clearly defining the LSTF is challenging. To address it, we will deeply discuss it from both relative and absolute perspectives.

From the relative perspective, the long prediction horizon task is usually defined as the LSTF. The long or short is a relative concept which is usually based on a comparison result. Currently, many LSTF definitions are based on the relative concept with comparison results [18,47] and no clear definitions are provided [33,47,48]. They elucidate the LSTF problem by comparing the prediction accuracy and inference performance of the model under different prediction horizons [17,49]. In longer prediction horizon tasks, the prediction

accuracy and inference performance tend to decrease compared to shorter prediction horizon tasks. Based on it, the prediction task with longer prediction horizon is usually defined as LSTF.

From the absolute perspective, the LSTF is usually derived from the application area [21], which varies with industry sectors. In most cases, the LSTF's definition is highly related to the cycle, which is a critical predictor for forecasting tasks [50]. The LSTF can be considered when the prediction range is close to or exceeds the maximum cycle of the data in the forecasting task. For instance, in finance, Queensland¹ and St.helena² has defined the forecasting tasks for over 10 years as the LSTF in the local Fiscal reports, while the cycle is usually between 2 to 10 years in the economic data [51]; in meteorology, in weather forecasting, the "Dictionary of Atmospheric Sciences" [52] defined the weather forecasting tasks with predict horizon over 1 month as the LSTF; the World Meteorological Organization (WMO)³ stipulated that the predict horizon of LSTF must be over 30 days and less than 2 years [53]. Nevertheless, the maximum cycle of the meteorological data in these forecasting tasks is usually a month or a quarter; in electrical load forecasting, Khuntia S. R. [54] and many real-world practices [55–58] termed the predict horizon of a few years (> 1 year) up to 10–20 years as the LSTF problem, where the maximum cycle of data is only up to 1 year; in traffic prediction, James J Q [59] and Peng H [60] categorized the tasks with predict horizon of several days as LSTF problems, while the maximum cycle as upper bounded by a few hours or 1 day; in blood glucose prediction, the LSTF was implemented as foretelling the at least 60 minutes, however, the maximum cycle is usually less than one hour.

Therefore, based on the investigation of real-world practices and research works in various industry sectors, it is naturally concluded that the LSTF is defined on the comparisons between predict horizons and cycles; In most of the tasks, if predict horizon is approaching or exceeding the maximal cycle, the forecasting task is viewed as a LSTF problem.

3. Review and classification of DL techniques for LSTF

In this section, we review the latest results in the field of DL for LSTF and classify the developed models based on the different technical routes of their effective architectures. Model feature extraction capability and architecture are crucial in LSTF. Therefore, based on the dominant feature extraction architectures used in the models, we can classify the models into RNN-based model, CNN-based model, GNN-based model, Transformer-based model, Compound model. For some special and novel methods that appear in the field and do not belong to the above categories, we classify them as miscellaneous methods. Finally, we review the evolution of the networks in the LSTF field in order to acquire the popular feature extraction architecture in LSTF.

¹ <https://www.legislation.qld.gov.au/browse/inforce>

² <https://www.cityofstheleena.org/finance/page/long-range-financial-forecast>

³ <https://community.wmo.int/en/global-producing-centres-long-range-forecasts>

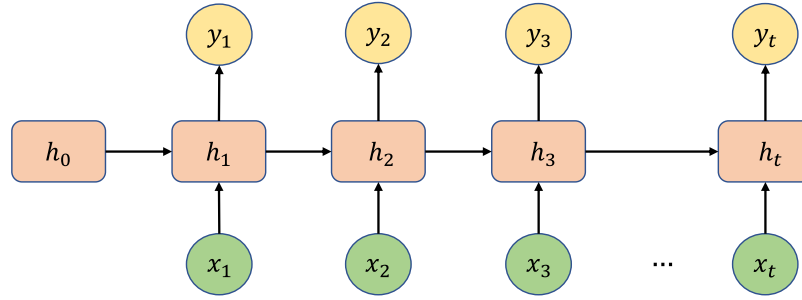


Fig. 8. Unrolled RNN Structure. At each time point t , the network is subjected to external input x_t and the hidden state h_{t-1} at the previous time point, while the hidden state of the network at t time point is updated to h_t , and y_t is the network output.

Note that the criterion for our collection of literature is whether the work enhances the predictive performance of the LSTF. For example whether it enhances the capture of long-term dependencies, or whether the specific task context of the study is long-term prediction.

3.1. RNN-based model

RNNs were first proposed by Elman et al. [61]. RNNs store past information by introducing state variables and use them together with the current input to determine the current output, and their innate structural designs are well suited for processing temporal information, the structure is shown in Fig. 8. Therefore, RNNs and their associated family of models have been widely utilized for processing sequential data. However, because vanilla RNNs suffer from gradient disappearance and gradient explosion in long sequences, Graves et al. [62] designed long short-term memory (LSTM), which greatly alleviated the problem of early RNN training by using gating units and memory mechanisms. Later, due to the computational complexity of LSTM, the large number of required computational resources and the slow training process experienced with a large amount of data, researchers [63] proposed an improved LSTM variant called a gated recurrent unit (GRU) model, which combines the forgetting gate and the input gate of LSTM into a single “update gate”, thus simplifying and reducing the model. This simplifies the model and reduces the number of model parameters, allowing it to converge faster during training. Table 3 summarizes and presents the recent works on RNN-based models for LSTF.

RNN variant. Vanilla LSTM has a strong ability to capture long-term dependence. Jung et al. [35] proposed an RNN model with LSTM (a monthly PV power generation prediction model) for predicting the amount of PV solar power that could be generated at a new location. The model uses an RNN with an LSTM layer for monthly data series (monthly solar radiation) to learn temporal and topographic variations in solar radiation and weather conditions to obtain temporal patterns observed at multiple sites. A multivariate time series forecasting method was developed based on bidirectional LSTM (BiLSTM) [73], and the model was trained using several features, such as the number of coronavirus detections, wind speed, ambient temperature, biomass, solar and wind energy supply, and historical electricity demand, to achieve satisfactory results. To improve the capture of long-term future dependencies, the winning entry of the M4 prediction competition, ES-LSTM [68], combines exponential smoothing (ES) with LSTM, where ES allows the method to efficiently capture the main components of individual sequences (for deseasonalization and normalization), while the LSTM network allows for nonlinear trends and cross-learning. This network is essentially hierarchical and utilizes both global (applied to a large subset of all sequences) and local (applied to each sequence individually) neural network parameters to enable cross-learning. Further, to addresses the problems of training long sequences in RNNs, such as capturing complex dependencies, vanishing and exploding gradients, and effective parallelization. The dilated RNN [64] constructs a model

for learning temporal dependencies at different scales and different levels. This is done combining multiple recursive expansion layers with hierarchical expansion, which is superimposed by multiresolution recursive expansion jump connections, for flexible combination with different RNN units. Effectively alleviates the three problems of RNNs network structure in LSTF.

Early work was mainly conducted by optimizing RNN networks themselves and combining them with traditional statistical methods. Since 2014, the emergence of structures such as attention mechanisms, sequence-to-sequence (Seq2Seq) structures and generative adversarial networks (GANs) has brought new directions regarding the optimization of RNN models in LSTF.

RNNs with Attention. In 2014, the attention mechanism was proposed and has since been widely used in the CV field, but in recent years, attention mechanisms has started to be combined with RNNs to capture more relevance from the data, which making it possible to predict the more distant future. An RNN model based on an attention mechanism and LSTM was designed by [72]. The model first uses a spatial-attention LSTM to extract the spatial correlations between multiple time series data, and after applying the attention mechanism to the whole time series, the spatial-attention LSTM is then used to extract the spatial relationships between the dependent variable time series and the independent variable time series to extract the event correlations. Achieving accurate LSTF for multivariate time series may not be easy because a portion of multiple independent variable time series data may not be helpful for prediction and may even impair the prediction accuracy, but this approach improves the resulting data quality by filtering the segmented multiple independent variable time series data through an attention mechanism to improve the prediction accuracy. Similarly to improve the ability to capture long-term correlations in the data. Y Xiang et al. [65] presented DA-RNN which is a two-stage RNN based on an attention mechanism. The model adaptively extracts correlations and hidden features among the input data at each time step through an attention mechanism in the first stage. In the second stage, the hidden states of the encoder are selected at all time steps by the temporal attention mechanism. Thus, the model adaptively extracts the most relevant features among the input data and captures the long-term temporal correlations seen in the input sequence. On the other hand, Sarkar et al. [74] similarly combined LSTM and attention mechanisms, thus proposing a spatio-temporal attention mechanism called STAM. The model uses an LSTM layer in the encoder to learn temporal correlations in the input data. At each output time step, the model first searches for the most relevant time step and the most important variable. The model then makes predictions based on the computed spatial and temporal context vectors.

RNNs with Seq2Seq. In addition to attention mechanisms, Seq2Seq has made a large splash in natural language processing (NLP) since 2014. Compared to the vanilla RNN, it effectively improves the ability to handle long sequences in multi-step forecasting. Fan et al. [67] borrowed the encoder-decoder structure to combine it with LSTM for LSTF. Their model uses a bidirectional LSTM decoder to first propagate future information in both the forward and reverse directions

Table 3

RNN-based Model. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
Dilated RNN [64]	2017	Dilated Connection+RNN	Vanilla RNN, Vanilla LSTM, VanillaGRU, Stacked RNN, Stack LSTM Stack GRU, Skip RNN	Dilated skip connection to addresses the problems of training long sequences in RNNs, 1. Capturing long-term and complex dependencies; 2. Vanishing and exploding gradients; 3. effective parallelization.	† ‡ §
DA-RNN [65]	2017	Attention+LSTM	ARIMA, NARX RNN, Encoder–Decoder, Attention RNN, Input-Attn-RNN	The two stages of Attention select the most relevant feature variables and hidden states in LSTM.	†
MQ-RNN [66]	2017	LSTM+ Encoder–Decoder+MLP	–	Multi horizon LSTM with encoder–decoder can cold-starts in real life LSTF.	† ‡
Fan et al. [67]	2019	LSTM+Encoder–Decoder+Atten	Gradient-Boosting, POS-RNN, MQ-RNN, TRMF	Attention to fuse multimodal features of LSTM.	‡
ES-LSTM [68]	2020	ES+LSTM	–	ES and LSTM capture linear and nonlinear relationships.	† ‡
Jung et al. [35]	2020	LSTM	–	Achieved long-term power generation forecast for PV plants by LSTM.	‡
MTSMFF [69]	2020	LSTM + Encoder–Decoder + Atten	SVR, RNN, CNN, LSTM, GRU, seq2seq, seq2seq-BI, seq2seq-ATT	Attention to select the hidden state of BiLSTM.	‡
LGnet [70]	2020	GAN+LSTM	LR, XGBoost, MICE, GRUI, GRU-D, BRITS	GAN to counter training and compensate for missing data in LSTM.	‡
ATFN [71]	2020	NN + S2S + GRU + DFT	S2S, S2SAttn, SFM, mLSTM, ND	DFT for periodicity, GRU for trending, NN to fuse multiple features.	‡ †
Yoshimi et al. [72]	2020	LSTM+Atten	LSTM, DSTP	DFT for periodicity, GRU for trending, NN to fuse multiple features.	‡
Liu et al. [73]	2021	BiLSTM	–	Long electricity demand forecast with the COVID19 impact.	‡
Gangopadhyay et al. [74]	2021	LSTM+Atten	SVR-RBF, Enc-Dec, LSTM-Att, DA-RNN	STAM to capture the most relevant variables at each time step.	‡

while taking the dynamic future information, such as promotions and calendar events, into account. Then, at each time step in the future, several different periods of history are focused on, and attention vectors are separately generated. The work treats different historical periods as different modalities and combines them by learning the relative importance of each modality for predicting the current time step. The study demonstrated that jointly learning temporal attention values for multiple historical periods and fusing them with multimodal attention weights can improve the prediction accuracy of the results to predict further into the future. The MQRNN [66] is a multihorizon prediction model that simultaneously predicts the values of multiple future time steps. MQRNN uses an encoder–decoder structure and changes the decoder part from an RNN to multiple full joins. The model directly stitches together the context of the last moment of the encoder and all future features through a full connection to generate the context of each moment and a global overall context, interweaving coarse and fine-grained features to improve the accuracy of predicting distant futures.

RNNs with Seq2Seq and Attention. The MTSMFF model [69] similarly combines LSTMs, encoder–decoder structures and attention mechanisms. It uses BiLSTM with temporal attention as an encoder network, adaptively learns long-term correlations and the implied correlation features of multivariate temporal data, and then decodes them with a stacked vanilla LSTM. Improves both the ability to handle long sequences and the ability to capture long-term correlations.

RNNs with GAN. Additionally, in 2014, the GAN was created as a revolution in the field of DL. The GAN was then introduced by Trirat et al. [70], who proposed a new framework called LGnet for solving the MTS prediction problem in the presence of missing values in multivariate temporal data (MTS). The framework constructs representative local features for each variable of the MTS problem and then uses an LGnet storage module to explicitly utilize the knowledge derived from external sequences to generate global estimates for the missing values. In addition, the modeling process of the global temporal distribution is enhanced by adversarial training. This work overcomes the case of missing values to ensure the validity of the prediction results, which is often affected by the growth of the prediction horizon

in LSTF, and this research overcomes the case of data with missing values to reduce the data integrity requirement in LSTF.

RNNs with Time-frequency Analysis Method. Time–frequency analysis can effectively indicate the relationship between revealed features rather than relying on model capture, and indicating the revealed feature relationship can effectively improve the accuracy to forecast distant future. Yang et al. [71] proposed an adaptive time–frequency network (ATFN) to solve the LSTF problem. The model first uses AugS2S2S, an augmented Seq2Seq model based on a GRU, to learn the trend features of complex nonstationary time series, then captures the complex periodic patterns of the time series using frequency domain blocks, and finally combines the trend and periodic features to produce the final forecast using a fully connected neural network. This study demonstrated the advantage of utilizing pure time domain models for LSTF problems involving weakly periodic time series, while frequency domain methods have significant advantages for LSTF problems with strongly periodic time series. Therefore, hybrid time domain–frequency domain models form a future direction for LSTF research.

3.2. CNN-based model

CNNs are very widely used structures in the field of computer vision. Its feature reduction and parallel processing capability in processing long sequences has led to attempts to apply it to LSTF. However, capturing short-term local dependence is a property of CNNs, and how to effectively capture long-term global dependencies in LSTF is of concern. It was not until causal convolution [75] and dilated causal convolution [76] were proposed that this problem was effectively alleviated, as shown in Fig. 9. In this section, we summarize and present the recent research works on CNN-based models in Table 4 and below.

CNN variant. TCN [77] is a classic model used for modeling and analyzing sequential data. It uses causal convolution to avoid information leakage and dilated causal convolution to handle cases with long sequential inputs. In addition, the above work introduced a residual joining approach to deepen the network. This demonstrates that a simple convolutional network structure can also perform as effectively as an RNN model such as LSTM when processing sequential data. And its

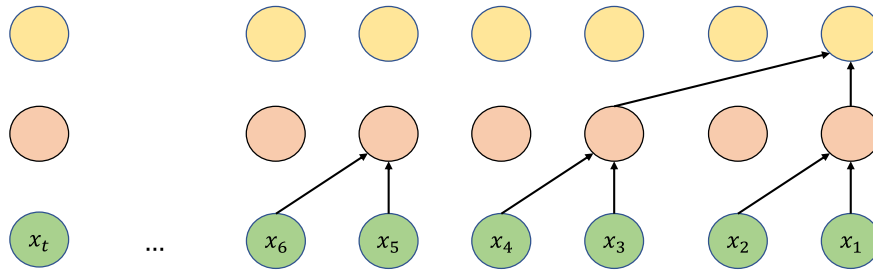


Fig. 9. The structure of CNN with two layers of causal convolutional layers. The input layer $Dilated = 1$, the hidden layer $Dilated = 2$.

Table 4

CNN-based Model. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
TCN [77]	2018	Causal CNN + Dilated CNN	LSTM, GRU, RNN	Dilated causal convolution can handle the long sequence inputs, residual connection to deepen the network.	‡
DSANet [78]	2019	Self-Atten+CNN+AR	VAR, LRidge, LSVR, GP, GRU, LSTNet-S, LSTNet-A, TPA	Parallel convolution for global and local time correlation, AR for robustness.	† ‡
PFNet [79]	2020	Highway-CNN+MLP	VAR, RNN (AR+GRU), MHA, LSTNet, MLCNN, MTGNN	LTPM for long-term trend, SFPM for short-term volatility.	† ‡
HyDCNN [80]	2021	Dilated CNN +AR	AR, VAR-MLP, GRU, TCN, LSTNet, TPA-LSTM, MTGNN	Autoregressive and dilated causal convolution capture cyclicity and trend, respectively.	‡
SCINet [81]	2022	CNN + Encoder-Decoder	Autoformer, Informer, Transformer, TCN, LSTNet, TPA-LSTM	A hierarchical downsample-convolve-interact framework to effectively models time series with complex temporal dynamics.	‡ §

Table 5

GNN-based Model. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
MTGNN [82]	2020	GCN+TCN+ResNet	AR, VAR-MLP, GP, RNN-GRU, LSTNet, TPA-LSTM, DCRNN, STGCN, Graph WaveNet, ST-MetaNet, GMAN, MRA-BGCN	The graph learning module for adaptive extraction of sparse graph adjacency matrix with TC and GC module.	† ‡
AutoSTG [83]	2021	GCN+Meta Learning	HA, GBRT, GAT-Seq2Seq, DCRNN, Graph WaveNet, ST-MetaNet+, RANDOM, DARTS	Meta learning to learn adjacency matrix and convolution kernel size for dynamic graph learning.	† ‡
DMST-GCN [84]	2021	Dynamic Graph Constructor+DGCN+TCN	HA, VAR, LR, XGBoost, DCRNN, ASTGCN, GMAN, Graph Wavenet, MTGNN	Dynamic graph constructor based on tensor decomposition.	† ‡
REST [85]	2021	DCRNN+GCN	ARIMA, VAR, SVR, FC-LSTM, WaveNet, DCRNN, Graph WaveNet	EINS to infer multimodal directed weighted graphs.	† ‡
TPGNN [86]	2022	GNN + Encoder-Decoder + Attention	VARMLP, GP, RNN-GRU, LSTNet, TPA-LSTM, MTGNN, TPGNN, Informer, Graph WaveNet, DCRNN, STGCN	A TPG module to represent the correlation as a time-varying matrix polynomial.	† ‡

parallelism and feature compression properties make it more suitable for LSTF than vanilla RNN. HyDCNN [80] also solved problems such as the inability of ordinary CNNs to capture long-term correlations. It is a hybrid framework based on dilated causal convolution, captures the nonlinear dynamics between sequences with dilated causal convolution and captures the sequence linear dependencies using autoregressive modules. In addition, HyDCNN consists of several hybrid modules, each of which targets a trend pattern or a kind of periodic pattern. To further capture periodic temporal patterns, a new jumping scheme was introduced in the hybrid modules. It captured multi-granularity cyclicity and trending to achieve improved forecast accuracy to predict further. Time series data have two characteristics, long-term trends and short-term fluctuations, which together constitute the complex structures of time series data. PFNet [79] was designed as a framework that can capture the long-term trends and short-term fluctuations of time series in parallel. The framework consists of three submodules: a long-term trend forecasting module (LTPM), a short-term fluctuation forecasting module (SFPM) and an information fusion module (IFM). The LTPM

uses Highway-CNN to capture long-term trend features, and the SFPM uses the differences between time series data as inputs. The Highway-CNN structure is also used for feature extraction, but this part extracts the short-term fluctuation features in the data and uses an MLP to do it. The IFM combines the features extracted by the two other modules to obtain the final prediction result. Compared with other baselines, this framework can more accurately capture the long change trends of time series to obtain higher prediction accuracy.

CNN with Attention. Although LSTM\GRU and other combinations with attention show better performance in capturing long-term predictions, they do not perform well on multivariate long-term forecasting. To optimize the long-term correlation capture problem for multivariate long-term prediction, Huang et al. [78] proposed a dual self-attentive network (DSANet) for involving dynamic periodic or aperiodic sequences. The model starts from both global and local temporal patterns, omits the recursive pattern of traditional RNNs, and employs two parallel convolutions (global temporal convolution and local temporal convolution) to capture a complex mixture of global and local temporal

patterns. In addition, the model still uses a self-attentive mechanism to learn the connections between multiple time series. To further improve its robustness, the model also integrates the traditional AR linear model in parallel with the aforementioned nonlinear neural network.

CNNs with Seq2Seq. Similarly, there have been attempts to combine CNNs and Seq2Seq to optimize long time series prediction problems. Liu et al. [81] out propose a SCINet with multiple convolutional filters to extract unique and valuable temporal features from downsampled subseries or features. By combining these rich features aggregated from multiple resolutions, SCINet effectively models temporal dynamics with complex multi-resolution and can better capture both short-term (local temporal dynamics) and long-term (trend, seasonality) temporal dependencies to make accurate forecasts in LSTF.

3.3. GNN-based model

The GNN is a network structure that is commonly used for spatio-temporal multivariate time series forecasting. Spatio-temporal data is a special kind of time series data. Unlike general time series data, spatial-time-series data exhibits temporality and spatiality, and there are dependencies between its features. Therefore, predicting such data requires models that can capture not only the temporal dynamic features of the data but also the spatial static features among the data. GNNs have developed rapidly in these two years by combining the ideas of graph theory with previous DL ideas, producing networks such as GCN, GAT, GAEs, GGNs, and GSTMs. Many existing works have also reviewed the current GNNs in terms of several aspects, such as their model structures and application areas [21,87–90]. Therefore, in this paper, we do not rediscuss them but only provide some complementary introduction to some challenges and work of GNNs on spatio-temporal long-term forecasting based on the previous work. Table 5 provides a summary of the discussed complementary GNN-based modules.

In LSTF, the predefined static graph structure is often not very effective at spatio-temporal long-term forecasting. As the prediction horizon increases, the hidden topological graph structure usually changes. And people started to research adaptive and dynamic graph structures to solve this problem of long-term spatio-temporal forecasting.

GNN with adaptive dynamic graph structure. As the predict horizon increases, the impact between nodes usually changes with traffic flow. This change in graph structure is especially significant in long-term traffic forecasts. So Han et al. [84] designed DMSTGCN based on DGNNs for this problem by using a dynamic graph convolution module to predict traffic speed. The model designs a dynamic graph constructor based on tensor decomposition for the dynamic spatial relationships between road segments while considering the periodic and dynamic characteristics of traffic; in addition, dynamic graph convolution is utilized to capture the spatial correlations among different road segments. This approach alleviates the problem of changing graph structure in long-term traffic forecasting through interpretable dynamic spatial relationships of road segments. In addition, other researchers have explored adaptive graph structure relationships to alleviate this problem. AutoSTG [83] was proposed as a new automatic prediction framework for spatio-temporal graphs, in which spatial graph convolution and temporal convolution operations are employed to capture complex spatio-temporal correlations. In particular, the framework uses meta-learning techniques to learn the adjacency matrix of the spatial graph convolution layer and the kernel of the temporal graph convolution layer from the meta-knowledge of the attribute graph. Similarly, MTGNN [82] was established as a general GNN structure. Instead of requiring a well-defined predetermined graph structure, this model uses a graph learning module to adaptively extract a sparse graph adjacency matrix based on the input data and then uses the extracted graph adjacency matrix to capture the spatial and temporal correlations of the data by interleaving a graph convolution module with a temporal convolution module. Furthermore, to avoid the vanishing gradient problem, residual and skip connections are added between the

temporal convolution module and the graph convolution module. High-quality graphs can improve the accuracy in spatio-temporal long-term prediction, and the authors who proposed REST similarly concluded that the effectiveness of GCN in spatio-temporal prediction is closely related to the quality of the graph structure. Moreover, REST [85] can automatically mine inferred multimodal directed weighted graphs. The model combines an inference edge network (EINS) and a GCNS. The EINS projects spatial dependencies between time series nodes from the temporal side to the spatial side and generates multimodal directed weighted graphs for the GCNS. From the temporal side to the spatial side, the GCNS uses these spatial correlations to make predictions and then introduces feedback to optimize the EINS. The spatial dependencies inferred by the EINS motivate the GCNS to make more accurate predictions, and the supervised training process of the GCNS helps the EINS learn better distance measurements. These two components collaborate with each other to build high-quality graphs, improving accuracy when making long-term spatio-temporal forecasting. Also for the dynamic graph problem in LSTF, Liu et al. [86] proposed TPGNN, which represents the correlations of dynamic variables as temporal matrix polynomials in two steps. First, we use a static matrix basis to capture the overall correlations. Then, we use a set of time-varying coefficients and matrix basis to construct matrix polynomials for each time step. All its long-term prediction aspects reach the state-of-the-art.

3.4. Transformer-based model

Google proposed the original Transformer [32], a model that completely discards traditional CNNs and RNNs, to handle sequence tasks in 2017. A Transformer captures correlations among sequence data mainly through a self-attention mechanism, and it has better parallelism than RNN structures for interacting with global information, as in CNNs. Compared with the traditional DL architecture, the self-attention mechanism in a Transformer is more interpretable and has epochal significance. However, Transformers also have some drawbacks; for example, their computational complexity is too large, and the time complexity of the self-attention computation is $O(n^2)$, leading to high computational costs. The utilization of location information is not obvious, the position embedding in the model embedding process is not effective, and long-distance information cannot be captured. In this work, the developed Transformer was mainly used for tasks in the NLP domain, such as machine translation. Due to the similarity of sequence prediction tasks, and its large model capacity and easy to model long time dependence, soon it were implemented into LSTF to the field of time series prediction, and its shortcomings were gradually improved. as shown in Table 6.

Transformer variant. Wu et al. [92] introduced the vanilla Transformer to the field of temporal prediction for influenza disease prediction. In temporal prediction, instead of processing data in an ordered sequential manner as done by an RNN, the Transformer processes the entire given data sequence and uses a self-attention mechanism to learn the dependencies in the temporal data. Therefore, the model relies heavily on embedding to obtain the positional relationships between sequences. However, the model often fails to adequately encode long sequences when processing long-sequence inputs and tends to lose long-term dependency items. To address the long-term dependency items of repeatable fluctuation patterns, Lin et al. [94] established a Transformer-based SpringNet for solar prediction. They proposed a DTW attention layer to capture the local correlations of time series data, and this attention mechanism can help capture repeatable fluctuation patterns and provide accurate predictions, especially in the presence of fluctuations. With the same purpose, Chu et al. [99] combines Autoformer, Informer and Reformer, proposed a forecasting model based on Stacking ensemble learning, where the results of the base learner are applied to train a meta-learner (MLP), which generates future forecasting structures from meta-learning. In contrast, TCCT [98] designs a CSPAttention module that merges CSPNet with

Table 6

Transformer-based Model. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
LogTrans [91]	2019	LogSparse self-atten + Transformer	ARIMA, ETS, TRMF, DeepAR, DeepState	Convolutional Self-Attention to reduced time complexity, demonstrated Transformer's ability to handle long-term dependencies.	‡ §
Wu et al. [92]	2020	Transformer	ARIMA, LSTM, Seq2Seq(GRU)+Attn	Transformer for Influenza LSTF.	‡
AST [93]	2020	GAN + Transformer	ARIMA, ETS, TRMF, DeepAR, DSSM, ConvTrans	Sparse attention with GAN to reduce error accumulation.	‡ §
SpringNet [94]	2020	Spring Attention + Transformer	DeepAR, LogSparse Transformer	Spring Attention for repeatable long-term dependency fluctuating patterns.	‡
Lee et al. [95]	2020	Partial Correlation-based Attention + Series-wise Multi-resolution Convolution + Transformer	–	Partial Correlation-based Attention for improving pair-wise comparisons-based attention disadvantages.	‡
Informer [17]	2021	ProbSparse Self-attention+ Self-attention Distilling + Generative Style Decoder	LogTrnas, Reformer, LSTMa, DeepAR, ARIMA, Prophet, LSTnet	Sparse and computationally effective transformer architecture.	‡ §
Autoformer [33]	2021	Sequence decomposition + Auto-Correlation + Transformer	Informer, LogTrans, Reformer, LSTNet, LSTM, TCN	Auto-Correlation and decomposition architecture.	† ‡ §
Pyraformer [96]	2022	Pyramidal Attention Module + Transformer	Informer, LogTrans, Longformer, Reformer, ETC	Pyramidal Attention Module for multi-resolution representation.	† ‡ §
FEDformer [97]	2022	Fourier Enhanced + Wavelet Enhanced + Transformer	Autoformer, Informer, LogTrans, Reformer	Time complexity reduction with frequency domain decomposition based on Autoformer architecture.	† ‡ §
TCCT [98]	2022	CNN + CSPNet Attention + Transformer	Informer, ARIMA, Prophet, LSTMa	CSPAttention reduces computational costs.	‡ §
Chu et al. [99]	2022	Autoformer + Informer + Reformer + MLP	Autoformer, Informer, Reformer, Transformer	Stacking ensemble learning based forecasting model incorporating multiple Transformer variants and meta-learner	‡
Li et al. [100]	2022	Transformer + Gate Mechanism	Informer, Transformer, ProbTrans	A generalizable memory-driven deformer plugin with progressive training.	‡
Quatformer [101]	2022	Transformer + learning-to-rotate attention + trend normalization	Autoformer, Informer, LogTrans, Reformer, LSTM, TCN	A Quaternion architecture with learning-to-rotate attention.	† ‡ §
Muformer [18]	2022	Multi-granularity attention + Transformer + Kullback–Leibler	Informer, LogTrans, Reformer, LSTMa, DeepAR, ARIMA, Prophet	Muformer architecture for multi-sensory domain feature enhancement and multi-headed attentional expression enhancement.	† ‡
Sepformer [102]	2022	Transformer + Discrete wavelet transform	Informer, LogTrans, Reformer, LSTMa, LSTnet	Sepformer architecture that combines discrete wavelet transforms to enhance feature extraction and reduce the computational complexity.	† ‡ §
Triformer [19]	2022	Transformer + Patch Attention	Reformer, LogTrans, StemGNN, AGCRN, Informer, Autoformer	A triangular, variable-specific attention architecture implements the capture of linear complexity and different temporal dynamic patterns of sequences.	‡ §

a self-attentive mechanism and replaces the typical convolutional layer with an expanded causal convolutional layer to modify the distillation operation employed by Informer to attain exponential receptive field growth. In addition, the model develops a penetration mechanism for stacking self-attentive blocks, which helps Transformer-like models obtain finer information at negligible additional computational costs. To mitigate the problem that the high time complexity of the self-attention computation in LSTF, leading to high computational costs, the LogSparse Transformer model [91] was designed to reduce the time complexity to $O(L(\log L)^2)$ and break the memory bottleneck. The model employs convolutional self-attention, which uses causal convolution to generate Q and K to enable better integration of local contexts into the attention mechanism. The LogSparse Transformer with a time complexity of $O(L(\log L)^2)$ was then proposed; this method only needs to compute the dot product of each cell in each layer with a complexity of $O(\log L)$. Thus, only $O(\log L)$ layers must be stacked for the model to be able to access all the information. The prediction accuracy achieved for fine-grained, long-term dependent time series can be improved in cases with limited memory budgets. The Informer [17] further reduces the computational complexity and memory usage of the attention mechanism with ProbSparse self-attention to efficiently replace regular self-attention and incurs a time complexity of $O(L\log L)$ and a memory usage of $O(L\log L)$. At the same time, the

proposed self-attention distillation operation significantly reduces the spatial complexity of the model computation process. This gives the model the ability to handle longer sequence inputs. In addition, the decoder part uses one-step output generation to prevent the cumulative propagation of errors. Lee et al. [95] established a partial correlation-based attention mechanism built on a Transformer for conducting partial correlation analyses of attention scores and applied it to multivariate time series forecasting. In addition, a data-driven serieswise multiresolution convolutional layer was designed to represent the input time series data for domain-agnostic learning. However, their paper did not provide an experimental validation of the proposed model and instead discussed the theory and innovations part of the model in greater detail. The effectiveness of this model in practical prediction applications has yet to be tested. Pyraformer [96], one of the latest works in the field developed in 2022, further breaks the limits of Transformers in terms of computational time complexity. It utilizes a pyramidal attention mechanism to form multiresolution representations of time series, the maximum length of the signal traversal path in Pyraformer is a constant with respect to the sequence length L , and the temporal and spatial complexity levels reach $O(L)$, thus enabling the effective capture of interdependencies in LSTF. Similar to Praformer, Cirstea et al. [19] also considered from that perspective and proposed a triangular, variable-specific attention architecture Triformer, which

achieves linear complexity through Patch Attention while proposing light-weight approach to enable variable-specific model parameters. Moreover, Li et al. [100] proposed a generalizable memory-driven Transformer, which integrates multiple time series features to drive the prediction procedure through a global-level memory component. In addition, an incremental approach is used to train our model by gradually introducing Bernoulli noise to improve its generalizability. This work proposes a plug-and-play plug-in that can improve the accuracy of LSTF, alleviating the problem of model prone to overfitting in LSTF.

Transformer with Time-frequency Analysis Method. Usually, serial decomposition methods such as STL are used as preprocessing steps for TSF. In contrast, the Autoformer [33] framework, which breaks away from the traditional idea of using sequence decomposition for preprocessing, designs a deep decomposition architecture that progressively decomposes time series into two aspects: periodicity and trends. This deep decomposition architecture embeds the sequence decomposition strategy into an encoder-decoder structure as an internal unit of Autoformer. During the prediction process, the model alternates between prediction result optimization and sequence decomposition; i.e., progressive decomposition is achieved by gradually separating the trend and period terms from the hidden variables. In addition, the model also establishes an auto-correlation mechanism based on stochastic process theory to replace the self-attention mechanism in the vanilla Transformer, realizing serieswise connections and an $O(L \log L)$ computational time complexity. FEDformer [97], also one of the latest works in this field in 2022, takes an alternative approach to optimize the vanilla Transformer from the frequency domain perspective. By adding a Fourier enhancement module and a wavelet enhancement module to the Transformer, FEDformer captures the significant structure in a time series through frequency domain mapping, which achieves $O(L)$ linear computational complexity and storage overhead by randomly selecting a fixed number of Fourier components and achieves substantially improved prediction accuracy while effectively capturing long-term dependencies. Also for the LSTF time complexity optimization problem, Chen et al. [101] proposed a Quatformer framework in which learning-to-rotate attention introduces learnable period and phase information to describe complex periodic patterns, trend normalization to model normalization of the sequence representation in the hidden layer, and decoupling of the LRA using global memory to achieve linear complexity without loss of prediction accuracy while effectively fitting the multi-periodic complex patterns in the LSTF. Also to address the higher computational complexity of Transformer in LSTF, limited capture of fine-grained features of data, and poor interpretability, Fan et al. [102] proposed Sepformer architecture to enhance feature extraction and reduce the computational complexity of the model to $O(CN)$. Among them, Sepformer uses independent networks to extract data stream features in parallel. SWformer and Mini-SWformer, by using discrete wavelet transform and separating high and low frequencies. The time complexity optimization in the above work is for the Transformer redundant information problem, Zeng et al. [18] proposed Muformer with the role of feature enhancement through the input multiple perceptual domains processing mechanism, while the multi-cornered attention head mechanism as well as the attention head pruning mechanism enhances the expression of multi-headed attention. It alleviates the problem of redundant information input in LSTF.

Transformer with GAN. In recent years, the idea of GANs has been widely studied in the field of imaging. Wu et al. [93] introduced GANs to the field of LSTF and designed a GAN-based adversarial sparse Transformer (AST) to solve the problem of not being able to predict long series due to error accumulation. The model achieves improved robustness and generalization by using a Transformer with sparse attention as the generator and then utilizing an adversary to alleviate the error accumulation when predicting multiple steps. The adversarial training process improves the robustness and generalization of the model. In addition, it regularizes the prediction model to improve its sequence-level capability.

3.5. Compound model

As can be seen above, different backbone of base models have their own advantages and disadvantages in the face of LSTF. For example, RNNs are good at capturing long-term macro dependencies of sequence data, but suffer from problems such as gradient explosion and slow inference speed. CNNs, on the other hand, are fast in parallel, but are better at capturing short-term local dependencies among sequence data, etc. By combining the strengths and weaknesses of each type of model, researchers have targeted specific forecasting problems and difficulties. The compound models categorized in this paper refer to models that combine the techniques of the previous classes of models described above in model backbone, as shown in Table 7.

RNNs+CNNs. RNN is better at capturing long-term macroscopic dependencies between sequence data but with slow inference speed, while CNN parallel fast is better at capturing short-term local dependencies between sequence data. People start to combine the advantages of both models to better solve the problem of long-time dependencies in LSTF. A classic concise and effective model among the developed hybrid models is LSTNet [105]. This model integrates CNN, RNN and AR techniques by using CNNs and RNNs to extract short-term local dependency patterns between variables and to discover long-term patterns for determining time series trends. In contrast, the traditional AR linear model is used to join the nonlinear neural network parts in parallel, making the nonlinear DL model more robust to time series with large data scale changes. In addition, LSTNet was designed with a new recursive structure, the recursive jump structure, which exploits the periodicity of the input time series signals and is able to capture very long time-dependent patterns. DAQFF [108] was proposed as a spatio-temporal prediction model combining the advantages of RNNs and CNNs, to solve the long-term spatio-temporal correlation capture problem in multivariate forecasting. The model addresses the dynamic, spatial-temporal and nonlinear characteristics of multivariate air quality time series by first extracting local trend features and spatial features through a one-dimensional convolutional CNN and learning spatio-temporal dependencies through BiLSTM. The spatio-temporal correlations in the given data are effectively captured by a simple model structure. The MLCNN [110] also integrates the CNN, RNN and AR techniques and was inspired by psychological construct-level theory to achieve capture long-term dependence by fusing prediction information from different future visions. The model first extracts the multilevel abstract representation among data with a CNN, then inputs the extracted features into shared LSTM and performs long-range and short-term future prediction. At the same time, the main LSTM is also set to perform feature extraction for the main task to make more accurate predictions. In this case, the shared LSTM works as the encoder of the main task features, while the main LSTM works as the decoder. In addition, the traditional AR linear model is combined with the nonlinear part of the neural network. Finally, the future vision aspects of the codecs are fused by such an improved encoder-decoder architecture.

To solve the problem of capturing extremely long term patterns while effectively adding other variable information in multivariate time series forecasting, MTNet [104] integrates CNN, RNN, and attention and mainly consists of a large memory component, three independent encoders, and a jointly trained AR component. The memory component is used to store long-term historical data, while the encoders encode the input and memory data into feature values. After calculating the similarity between the input data and the data in the memory component, an attention mechanism is also used to calculate the attention weights of all memory blocks. In addition, the model borrows from the LSTNet structure by combining a traditional AR linear model with a nonlinear neural network to improve the sensitivity of the model to data scale changes. Zhao et al. [106] designed a new model to capture long-term dependence from a frequency domain perspective. Their work considered the fact that real-life time series are driven by multiple

Table 7

Compound Model. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
DCRNN [103]	2017	Diffusion Conv + RNN + Encoder–Decoder + Graph	HA, ARIMA, VAR, SVR, FNN, FC-LSTM	New form of convolution to optimize RNNs to capture spatial correlation using bi-directional random wandering on the graph.	‡ †
MTNet [104]	2018	CNN + Atten + RNN + AR	AR, LRidge, LSVR, GP, VAR-MLP, RNN-GRU, DA-RNN, LSTNet	Large memory component with CNN, RNN, Attention to storage of long-term historical data.	‡ †
LSTNet [105]	2018	GRU + CNN + AR	AR, LRidge, LSVAR, TRMF, GP, VAR-MLP, RNN-GRU	CNN for short-term local dependencies, RNN for long-term trend dependencies, AR for robustness improvement.	† ‡
Zhao et al. [106]	2018	Wavelet Transform + CNN + LSTM + Atten	ARIMA, SVR, ANN, LSTM, CNN, Ensemble of LSTM & CNN	CNN and LSTM for long and short term dependence capture after frequency domain change.	‡ †
RESTFul [107]	2018	GNN + CNN + GRU + MLP	SVR, ARIMA, MLP, GRU, Deep-GRU, Dipole, DA-RNN	Multi-resolution forecasting framework with CNN and GRU alternation.	‡ †
DAQFF [108]	2019	CNN+LSTM	ARIMA, SVR-POLY, SVR-RBF, SVR-LINEAR, LSTM, GRU, CNN, RNN	CNN for local trend features, LSTM for long-term spatial–temporal dependence.	‡
GC-LSTM [109]	2019	GCN+LSTM	MLR, FNN, LSTM	GCN for spatial correlation to assist LSTM long-term correlation capture.	‡ †
MLCNN [110]	2020	CNN + RNN + AR	VAR, RNN-LSTM, MTCNN, AECRNN, LSTNet	CNN combined with LSTM for multi-resolution feature extraction, AR for improving robustness.	‡ †
Forecaster [111]	2020	Markov + graph + Transformer	VAR, DCRNN, Transformer(same width/best width)	Gauss Markov random field theory to simplify transformer.	‡ † §
STNN [112]	2020	Transformer + GCN	HA, ARIMA, LSVR, FNN, FC-LSTM, STGCN, DRCNN, Graph WaveNet	Spatial-transformer and temporal-transformer to capture directed spatial dependence and long-term temporal dependence.	‡ † §
DeepFEC [36]	2021	ResNet + BiLSTM	MR, SVR, ANNs, RNN, GRU, LSTM, Bi-LSTM, DCNN, ST-ResNet, LC-RNN	Dynamic aggregation of spatial and temporal correlations.	‡
NAST [113]	2021	Spatial–temporal Attention+ Transformer	RNN-ED, TRANSFORMER, TST-NOLOGSPARSE	New spatial–temporal attention and non-autoregressive architectures.	‡ §
NET3 [114]	2021	TGCN+TRNN	DynaMMo, MLDS, DCRNN, STGCN	TGCN and TRNN collaboration improves long-term dependent capture and reduces the number of model parameters.	‡ † §
GCLSTM + GCTrafo [115]	2021	GCN + LSTM + Transformer	STAR, STCNN, EDLSTM, SVR	GCLSTM and GCTrafo for long-term dependent capture and reduction of the number of participants.	‡ † §
MS-LSTM [116]	2022	LSTM + Attention + Seq2Seq + CNN	AR, DeepAR, Transformer, LSTM, LST-Attn	MS-LSTM structure for LSTM with multi-scale feature extraction.	‡
ST-KMRN [117]	2022	GNN + CNN + Koopman + Encoder–Decoder + Self-attention	HA, Static, GRU, Informer, Graph WaveNet, MTGNN, KoopmanAE	A multi-resolution long-time feature modeling framework ST-KMRN incorporating self-attention and Koopman theory for physical information modules.	† ‡

potential components that occur at different frequencies. Therefore, the researchers drew on signal processing to explicitly reveal the frequency domain information in the input data using a wavelet transform and then used a CNN and LSTM based on the transformed data to capture local time–frequency features and global long-term trends, respectively. In addition, attention mechanisms were further used to distinguish the importance levels of different local features and to fuse the local features with global features. DeepFEC [36] was established as a new deep spatio-temporal contextual association network for predicting the energy consumption levels of roads. This approach uses ResNet to capture the spatial correlations between different road segments, while BiLSTM learns the temporal correlations between sequential data. The method achieves a decrease in the number of parameters and an increase in accuracy, which improves the processing of long series data and the ability to capture correlations between data. While Wang et al. [116] proposed MS-LSTM, a long time sequence prediction method using multi-scale feature extraction and sequence-sequence (seq2seq) attention mechanism under long short-term memory (LSTM) hidden state, using LSTM and CNN to extract long-term and short-term features at different scales, respectively, and using multi-scale information to assist long-term forecasting.

RNNs+CNNs+GNNs. Graph neural networks are also often fused with models such as CNN and RNN to improve the accuracy of long-term forecasting of the future by being used to capture spatial correlations between serial data. Qi et al. [109] combined the advantages of a GCN and an LSTM network to design a GCLSTM model and used it for meteorological data prediction. The model first introduces a GCN to capture the spatial correlations between different meteorological stations and then captures the temporal correlations among the observations at different moments via LSTM to finally achieve effective LSTF for meteorological data. Similarly, many researchers have started to introduce variants of GCN, RNN and CNN models for hybrid model experiments. The DCRNN [103] was designed as another new convolutional network form for optimizing RNNs, alleviating the LSTF problem in traffic flow forecasting. It uses bidirectional random wandering on the input graph to capture spatial correlations and uses a codec structure and scheduling sampling to capture temporal correlations. The relationships between traffic sensors are modeled by directed graphs, the traffic flow dynamics are modeled as a diffusion process, and a diffusion convolution operation is utilized to capture spatial correlations. Jing et al. [114] proposed a new tensor time series network model called NET3. The model consists of two modules: a tensor GCN (TGCN) and a tensor RNN (TRNN), where the TGCN captures the relationships between multiple graphs by generalizing the GCN from planar graphs to

tensor graphs, and the TRNN models the implicit dependencies between time series using tensor decomposition. In this way, the noise in the data and the number of parameters in the model can be effectively reduced. Improved the ability to handle long sequence data. REST-Ful [107] was then established as a multi-resolution framework that uses time series data at different granularities. The framework starts with a recursive framework (a GRU) to encode the temporal patterns at each resolution. A convolutional framework (a CNN) is used to fuse the temporal patterns of the data with different resolutions captured by the GRU. Finally, the learned feature vectors are fed into an MLP to obtain the prediction output. It provides more feature information for long-term prediction through multi-resolution feature representation.

Transformers+GNNs. Transformer shows better performance in LSTF, and people explore combining Transformer and GNN to solve the long-time prediction problem in spatio-temporal forecasting. The STNN [112] is a new spatio-temporal Transformer network that improves the accuracy of long-term traffic flow prediction by combining a Transformer and a GCN to dynamically model directional spatial dependencies and long-term temporal dependencies; the network is mainly divided into a spatial Transformer and a temporal Transformer. The spatial Transformer can be used to dynamically model the directional spatial dependencies between data by using a self-attentive mechanism to obtain the real-time status and direction of a traffic flow and to capture different spatial dependencies by using a multi-headed attention mechanism. The temporal Transformer, on the other hand, captures remote bidirectional temporal dependencies across multiple time steps. This reduces the accumulation of prediction errors and enables parallel training and prediction. Chen et al. [113] established a Transformer-based NAST model for spatio-temporal forecasting problems. This model was the first non-AR Transformer architecture and is used to overcome the output delay and cumulative error of the decoder in the traditional Transformer. The model has QGB as the core module; QGB generates queries in one step, and these queries have the same length as the target sequence. Then, by using the generated query Q, the time-transformed decoder is able to execute the prediction process in parallel, thus preventing error accumulation. In addition, the model utilizes a new spatio-temporal attention mechanism that first embeds the input features into a high-dimensional tensor and predicts attention maps in both the spatial and temporal directions. Then, the temporal attention map is permuted to obtain a temporal influence map, and the temporal influence map is used to perform the product operation with the spatial attention map to obtain the overall spatio-temporal attention map for joint learning so that the spatial and temporal dependencies can be processed completely. From another perspective, Forecaster [111], a graph-based Transformer framework for spatio-temporal data prediction, has also been proposed. The framework first learns the topology of the given graph through Markov random field theory and then replaces the linear layer of the Transformer with a sparse linear layer through this dependency graph structure, allowing the sparse Transformer to simultaneously capture the potential spatio-temporal dependencies, nonsmoothness and heterogeneity in the data and reducing the computational complexity of the Transformer to some extent. It effectively mitigates the impact of non-stationarity problems in LSTF.

Transformers+RNNs+GNNs. Moreover, Carrillo et al. [115] established a graph convolutional LSTM (GCLSTM) model and a graph convolutional Transformer (GCTrafo) model by fusing the ideas of LSTM, GCNs, and Transformers. Among them, the GCLSTM uses a recursive structure to dynamically combine the GCN and LSTM. GCTrafo, on the other hand, embeds a GCN into the Transformer framework for spatio-temporal dependency capturing. It effectively achieves spatio-temporal long-term prediction.

3.6. Miscellaneous

In addition to the above mentioned baseline models and the associated compound methods, researchers have explored the possibility of using other techniques in a nontraditional sense for LSTF, as shown in Table 8. In this paper, only some interesting techniques are characterized and explored in LSTF. For example, Cui et al. [46] proposed the idea that it is wrong to ignore or underestimate one of the most natural and fundamental temporal properties of time series, historical inertia (HI), in the LSTF problem. Therefore, they explored the development of a new baseline LSTF model called the HI model, which uses the most recent historical data point in the input time series as the prediction result. The final results showed that their method has an extremely strong ability to be compared with the SOTA models, proving its validity. Similarly, Li et al. [119] proposed the use of a simpler statistical method as a benchmark, proposing SNaive to serve as an effective nonparametric method to alleviate the difficulty of consistently low prediction errors in LSTF. And to address the problem, Zhou et al. [120] argued that this LSTF problem is caused by overfitting the noise presented in the history, so a frequency-improved Legendre Memory Model (FiLM) architecture was designed, which applies Legendre polynomial projections to approximate the historical information, uses Fourier projections to remove the noise, and adds a low-rank approximation to speed up the computation. It better alleviates the over-fitting noise in LSTF while speeding up the inference.

Moreover, meta-learning is widely used in major fields as a popular technique, and some of the works presented above also introduced meta-learning to assist with the model prediction process. [118] proposed a generic meta-learning framework; this approach can source time series datasets on which neural networks can be trained, and these networks can then be deployed on different target time series datasets without retraining. This solves the problem that the number of available historical samples for some target time series is too small, and therefore, it alleviated the problem of insufficient amount of historical data in LSTF.

Oreshkin et al. [121] investigated how to perform complex timing prediction tasks by means of simple interpretable strong DL models, proposing an N-Beats model based on MLP stacking and residual linking. Their subsequent work N-HiTs [49], improves N-Beats on the LSTF task by incorporating novel hierarchical interpolation and multi-rate data sampling techniques to emphasize components with different frequencies and scales while decomposing the input signal and synthesizing predictions.

3.7. Evolution of networks in LSTF

In this subsection, we summarize the number of the popularly feature extraction network structures used in LSTF by timeline since 2017, and try to summarize the trends and problems of popular feature extraction structures in recent years from them, which are shown in Fig. 10. A more fine-grained review graph is shown in Appendix B.

We can see that in 2017, people paid more attention to research involving RNN-based models; at this time, the time series prediction task was still inseparable from RNNs, and researchers were more inclined to encoder-decoder architectures. However, due to the natural cyclic AR structures of RNNs, they have problems such as the ease of gradient disappearance and gradient explosion as the network deepens, and a shallow network often does not sufficiently exploit data dependency when dealing with long sequences of data, so people have also started to explore other mechanisms such as convolution for time series applications.

The TCN relying only on dilated CNNs emerged in 2018, and traditional CNN are generally considered less suitable for modeling time-series problems. This mainly due to their convolutional kernel size limitations, which cannot effectively capture long-term dependence information. However, the unique convolutional structures in TCNs avoid

Table 8

Miscellaneous Methods. Remark denotes the research directions of each work in enhancing LSTF. †: Explicitly revealing the correlations between variables; ‡: Implicitly increasing the intersequence dependency learning abilities of models; §: Few parameters and low computational complexity.

Ref	Year	Technique	Baseline	Comments	Remark
HI [46]	2021	HI	Prophet, ARIMA, DeepAR, LSTMa, Reformer, LogTrans, Informer, Informer-	Simple historical inertia HI as a baseline model for LSTF.	† §
Oreshkin et al. [118]	2021	Meta learning	–	Only meta-learning enables prediction, solving available historical samples too few in LSTF.	‡
N-HiTs [49]	2022	MLP + Residual Connection	N-BEATS, FEDformer, Autoformer, Informer, LogTrans, Reformer, DiLRNN, ARIMA	A novel way of hierarchically synchronizing the rate of input sampling with the scale of output interpolation across blocks.	‡ §
SNaive [119]	2022	Linear Regression	LSTNet, LSTMa, Reformer, LogTrans, Informer, TCN, SCINet	The SNaive is an effective non-parametric method to achieve a lower error limit than deep learning methods.	† §
FILM [120]	2022	Legendre projection + Fourier analysis	FEDformer, Autoformer, S4, Informer, LogTrans, Reformer	A Frequency-Improved Legendre Memory Model (FILM) Architecture	† §

the problems of easy gradient disappearance and gradient explosion experienced by RNNs and achieve excellent results. Compared with those of RNNs. The parallelized processing, flexible fields and lower memory levels of TCNs have achieved good results, and researchers have started to study the use of CNNs combined with RNNs to build compound models. However, when processing long-sequence data, TCNs are still limited by the use of extended convolution to expand their perceptual fields; thus, they still suffer from the problem that their perceptual fields are not large enough, leading to poor model results.

The inherent sequential properties of RNNs hinder the parallelization of training samples, the perceptual fields of CNNs limits their ability to capture global dependencies, and an attention mechanism enables the dependencies of input-output sequences to be modeled without considering their distances in the input sequence. Therefore, in 2019, researchers started to explore the applications of Transformers in time series prediction. One such proposed method was LogTrans, and its excellent global dependency capture ability achieved good prediction results, verifying the effectiveness of Transformers model in LSTF. Furthermore, GNNs have also started to demonstrate their effectiveness in multivariate time series forecasting.

From 2020–2021, much research on Transformer-based models, GNN-based models, and compound models explored popularly. Transformers have the following drawbacks with respect to time series prediction: (1) $O(L^2)$ time complexity; (2) inadequate location information encoding and representation; and (3) inferior information acquisition to RNNs and CNNs. Researchers of GNN-based models prefer to study how to construct a suitable graph adjacency matrix for enhancing multivariate prediction. Until 2022, LSTF studies preferred Transformer-based models, whose excellent long-term dependency capture capabilities have yielded good results in LSTF. In addition, Transformer-based models also have some inherent drawbacks, and using compound models to complement each other is also a good solution.

4. Key problems and effective solutions of DL in LSTF

In Tables 3–7, we summarize the research directions of each work in enhancing LSTF. Based on above review, in this section we summarize the key problems encountered by the above work in LSTF. Note that these problems may not be unique to LSTF, but LSTF will exacerbate them. We categorized them into three points. In Section 4.1, we present the key problems encountered by LSTF in three directions. In Sections 4.2 and 4.3, we summarize the solutions and effective components provided in the current works for the first two key problems, respectively. In Section 4.4, we present our LSTF performance evaluation approach for the last key problem. The key LSTF problems and the effective solutions presented in this part are shown in Fig. 11.

4.1. The key problems of LSTF

LSTF is a challenging problem in both theory and practice. Below, we summarize the key problems encountered in LSTF in terms of three aspects.

- (1) **More and longer-term dependency.** “Capture dependency” refers to the ability of a forecasting model to capture the dependencies and patterns that exist within the time series data [122,123]. In other words, it refers to the model’s ability to accurately capture the temporal relationships between past and future observations in the time series [78]. Long sequence time series exhibit long-term dependencies, where the future values depend on the past values that are far away in time [100,116,119]. It poses the challenge to the model’s ability to capture dependencies [17]. Moreover, LSTF is characterized by inherent cyclicity and non-stationarity [124]. Thus, LSTF models need to learn a mixture of the short-term and long-term repeated patterns in the given time series [33]. An example is the hourly occupancy rate of a highway. Clearly, there are two repeating patterns: daily and weekly patterns. The former depicts the morning peak versus the evening peak, and the latter reflects the weekday and weekend patterns. An effective model should capture these two repeating patterns to make accurate LSTF. Therefore, this imposes more stringent requirements on the forecasting model in terms of learning of dependency.
- (2) **Computational complexity and Memory requirements.** Long sequence time series forecasting requires processing a large amount of data [101], which can be computationally intensive [33]. This can result in longer training times and require more computing resources [18]. Therefore, low computational complexity is very critical for LSTF. In addition, storing the entire sequence in memory can be challenging due to the limited memory available in computers [96]. Some models, such as recurrent neural networks, require a large amount of memory to process long sequences [64]. This can limit the length of the time series that can be used for forecasting.
- (3) **Performance evaluation method for LSTF.** When the prediction horizon grows, the length of the model prediction increases, requiring the model’s prediction results not only to be close to the true values but also to match the distribution of the true data. In addition, due to the growth of the prediction horizon, it is difficult to judge which model has better LSTF performance when the difference between the predictions of two models is very small, e.g., when the mean squared errors (MSE) of the two predictions differ by 0.001. For example, we cannot compare the superior results [42] of PatchTST/42 and DLinear for MSE=0.202 and MSE=0.203. We need to prove the superiority of one model over the other in a statistical sense. However, there

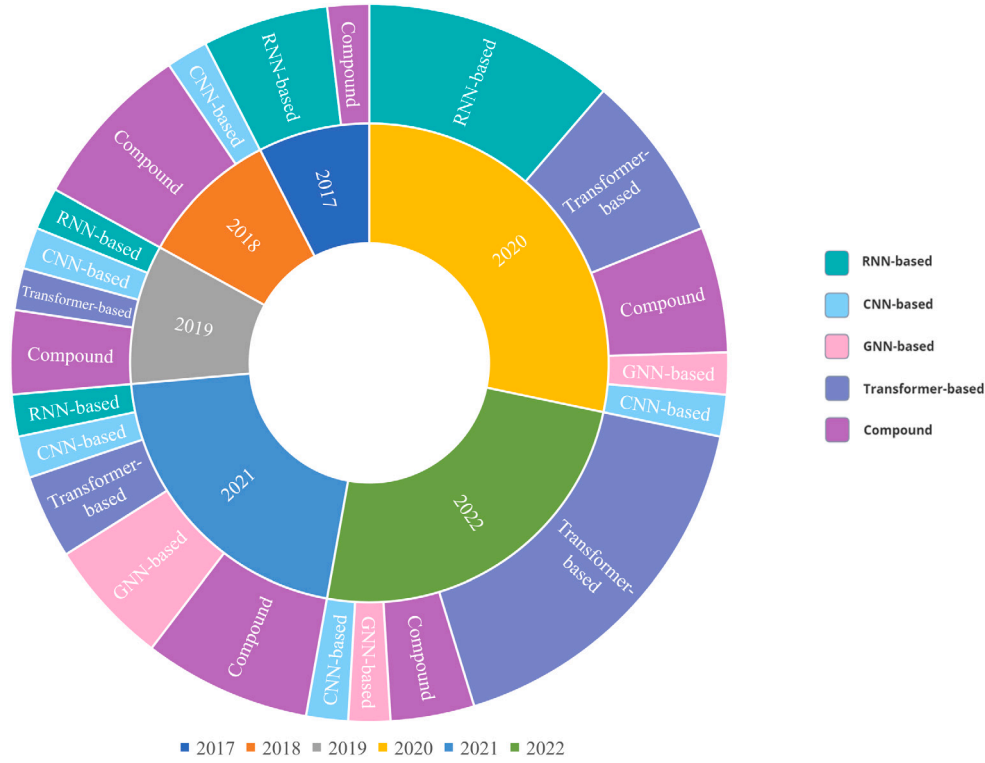


Fig. 10. Literature Statistics. Statistics on the number of research works on mainstream feature extraction network structures based on timeline.

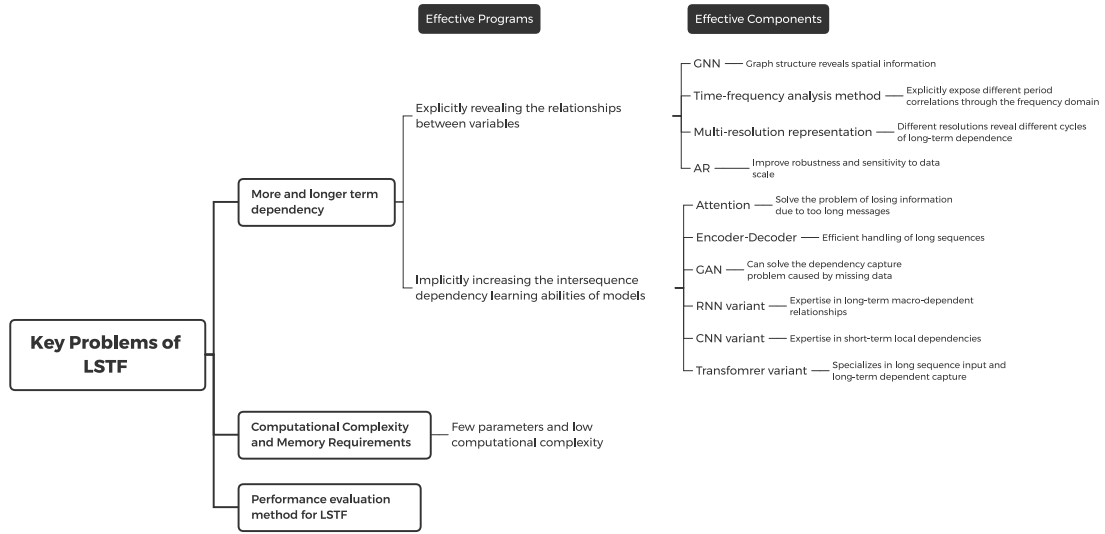


Fig. 11. A Summary of the Key Problems and Solutions in LSTF.

is no relevant performance evaluation. Therefore, a key problem involves systematically evaluating the LSTF performance of a model. In this paper, we apply the Kruskal–Wallis test to evaluate the LSTF performance of a model from two perspectives: model-model and model-self.

Now, we proceed to the possible solutions of LSTF based on the key problems listed above, and the details are as follows.

4.2. More and longer term dependency

Most of the existing work on LSTF has concentrated on how to achieve improved prediction accuracy by capturing more and longer-term dependencies between sequences. The majority of these studies

commence by explicitly exposing the relationships between variables and implicitly increasing the ability of the model to capture serial correlations.

4.2.1. Methods for explicitly revealing the relationships between variables

Previously, feature engineering was used to explicitly expose the relationships between variables by manually constructing complex features to reveal the correlations between serial data. In recent years, many novel techniques have also emerged in LSTF to explain the correlations between variables from other perspectives. GNNs have made a splash in the field of TSF in the last two years by modeling spatial correlations among data through graph structures as a way to capture more and longer-term dependencies. AutoSTG [83] automatically searches

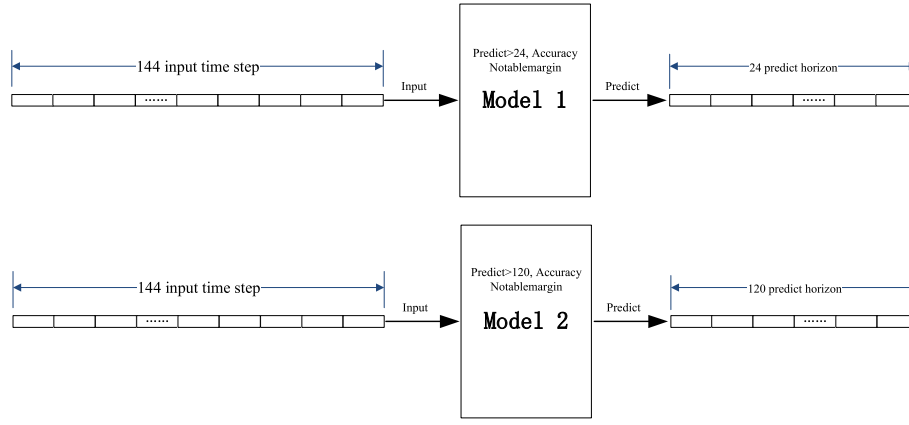


Fig. 12. Model-Self.

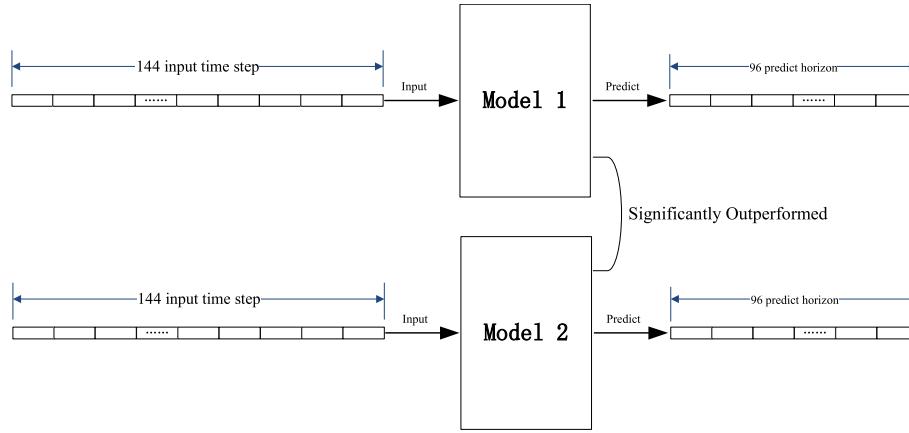


Fig. 13. Model-Model.

for suitable graph structures through a kind of meta-learning strategy, MTGNN [82] adaptively extracts sparse graph adjacency matrices based on data through a graph learning module, DGNNs [84] use dynamic graph structures to model spatial correlations, and CGM [149] even introduces multi-view information to model the relationships among variables more significantly by constructing a multi-view information-based heterogeneous graph. REST [85], on the other hand, exposes the linkages between variables by means of a multimodal directed weighted graph.

Time-frequency analysis methods are also effective techniques, and the ATFN has [71] demonstrated the significant advantage of frequency domain methods for the LSTF of strongly periodic time series through a time-frequency adaptive network. FEDformer [97] similarly starts from the frequency domain and uses Fourier variation and wavelet transforms to expose potential periodic information in the input data. Zhao et al. [106] also explicitly revealed the frequency domain information in the given data via the wavelet transform before further learning the long-term correlations between the series with their model.

The multi-granular data input approach reveals the relationships between variables in terms of amounts of information, and different granularities reflect the relevance levels of the different cycles of a feature. Lee et al. [95] used multi-resolution convolution to extract feature information at different granularities, Pyraformer [96] uses a pyramidal attention module for multi-resolution time series representation, and the MLCNN [110] similarly extracts multilevel abstraction representations from data to aid the prediction process. HTML [15], on the other hand, uses multiple views of a multi-source data input to reveal potentially complex long-term correlations among the same features from different dimensions.

4.2.2. Methods for implicitly increasing the intersequence dependency learning abilities of models

How to effectively increase the abilities of models to learn long-term correlations between sequences has been a hot research topic, and DL studies in this area can be traced back to LSTM, which was proposed to solve the memory problem of RNNs when dealing with long sequences. DL has evolved since then, and many efficient and effective components that can enhance the learning abilities of models have emerged. Jung et al. [35] performed LSTF for photovoltaic power generation cases by using vanilla LSTM. The dilated RNN [110] further mitigates the difficulty faced by RNNs when capturing complex correlations in long sequences by jump linking.

Attention mechanisms solve the information loss problem caused by excessive information length via screening, which can effectively improve the learning ability of the utilized model for long series data. To solve the problems of data redundancy and noise in multivariate time series LSTF, Yoshimi and Eguchi [72] combined an attention mechanism and LSTM to improve the quality of the input data by screening and segmenting multiple independent variable time series data through the attention mechanism. The DA-RNN [65] more effectively captures long-term event correlations by selecting hidden states at all time steps through a two-stage RNN based on an attention mechanism. Gangopadhyay et al. [74] also combined LSTM and an attention mechanism and proposed a spatio-temporal attention mechanism (STAM) to improve the LSTF capability of their model based on the computed spatial and temporal context vectors.

Encoder-decoder structures are also effective for enhancing the LSTF capability of a model, as they can efficiently handle long sequences while achieving sequence dimensionality reduction. Fan et al. [67]

Table 9
Evaluation measures.

Category	Performance measure	Remark	Ref
Scale-dependent	Mean Absolute Error (MAE)	Highlight the outliers even more	[11,33,36,46,71,78,82–85,104,107,109,125–132]
	Mean Square Error (MSE)	Expected value reflecting the square between predicted and truth values	[15,33,35,46,72,106,133]
	Largest Absolute Error (LAE)	Reflects only the largest error in the results and is susceptible to outliers and does not reflect the overall fit of the results	[134–137]
	Root Mean Square Error (RMSE)	For cases where the error is not very obvious	[11,27,35,36,70,71,73,82–85,92,104,107,109,111,125–132,138–141]
	Mean Squared Logarithmic Error (MSLE)	Only care about the relative between the true and predicted values, reflecting the percentage difference between them	[73]
	Mean Absolute Log-scaled Error (MALE)	–	[142]
	Root Mean Squared Log Error (RMSLE)	Significantly more penalties are imposed for under-predicted than over-predicted	[85,128,142]
Scale-independent	Mean Absolute Percentage Error (MAPE)	Using absolute values to avoid positive and negative errors canceling each other out	[27,35,36,70,71,82,84,85,111,127,128,131,133,140,141]
	IA	A non-dimensional, bounded measure of the degree of model prediction error	[109]
	Root Relative Squared Error (RRSE)	Reflects the mean offset between the target and the true regression line and is used to estimate the standard deviation of the residuals	[78–80,82,104,105,143,144]
	R-Squared (R^2)	Intuitively reflect the fit of the true and predicted values as a percentage	[35,126]
	Symmetric Mean Absolute Percentage Error (sMAPE)	Corrects for the asymmetric distribution characteristic of MAPE, but is unstable when both the true and predicted values are very close to zero	[85,121,138,145,146]
	Normalized Root Mean-square Error (NRMSE)	Normalize the value of RMSE to the (0,1) interval	[28,35,115]
	Root Mean Square Percentage Error (RMSPE)	Reflects the error rate and avoids the effect of the difference in magnitude between the true values on the error, when the predicted value is more than twice the true value, the RMSPE value may be so large that the evaluation is lost	[107,147]
	Relative Absolute Error (RAE)	expressed as a ratio, comparing a mean error (residual) to errors produced by a trivial or naive model	[79,80]
Scaled-error	Normalized Mean Absolute Error (NMAE)	Normalize the MAE	[115]
	Bean Absolute Scaled Error (MASE)	By comparing the prediction results with the output of a naive prediction method, it is susceptible to outliers and does not visualize the error in the prediction results	[145,146]
	Median Absolute Scaled Error (MdASE)	More robust compared to MASE, but cannot visualize the error of prediction results	[148]

combined an encoder–decoder structure with LSTM to jointly learn temporal attention values from multiple historical periods and fuse them with multimodal attention weights to improve the prediction accuracy of LSTF. The MQRNN [66] is a multihorizon prediction model with an encoder–decoder structure that can cold-start in real-life long-sequence forecasting problems. Notably, there are researchers who have combine attention mechanisms and encoder–decoder structures. MTSMFF [69] adaptively learns the long-term correlations and implied correlation features of multivariate temporal data through of temporal attention in BiLSTM as an encoder network.

GANs can solve some of the missing data problems encountered in LSTF. The authors of LGnet [70] introduced a GAN to TSF to overcome

missing value cases to ensure the validity of LSTF, which often results in the growth of the input size due to the increased prediction horizon. This work reduced the requirement of data integrity in LSTF.

Researchers have also started to explore the possibilities of CNNs in LSTF. The TCN [77] was proposed with a causal convolution to prevent information leakage, and the network uses dilated causal convolution to handle long-sequence inputs. HyDCNN [80] utilizes a hybrid framework based on dilated causal convolution to solve the problem of long-term dependence between ordinary CNNs that cannot fire sequences. PFNet [79], on the other hand, improves the LSTF capability of the associated model by capturing long-term trend features and short-term fluctuation features through its Highway-CNN

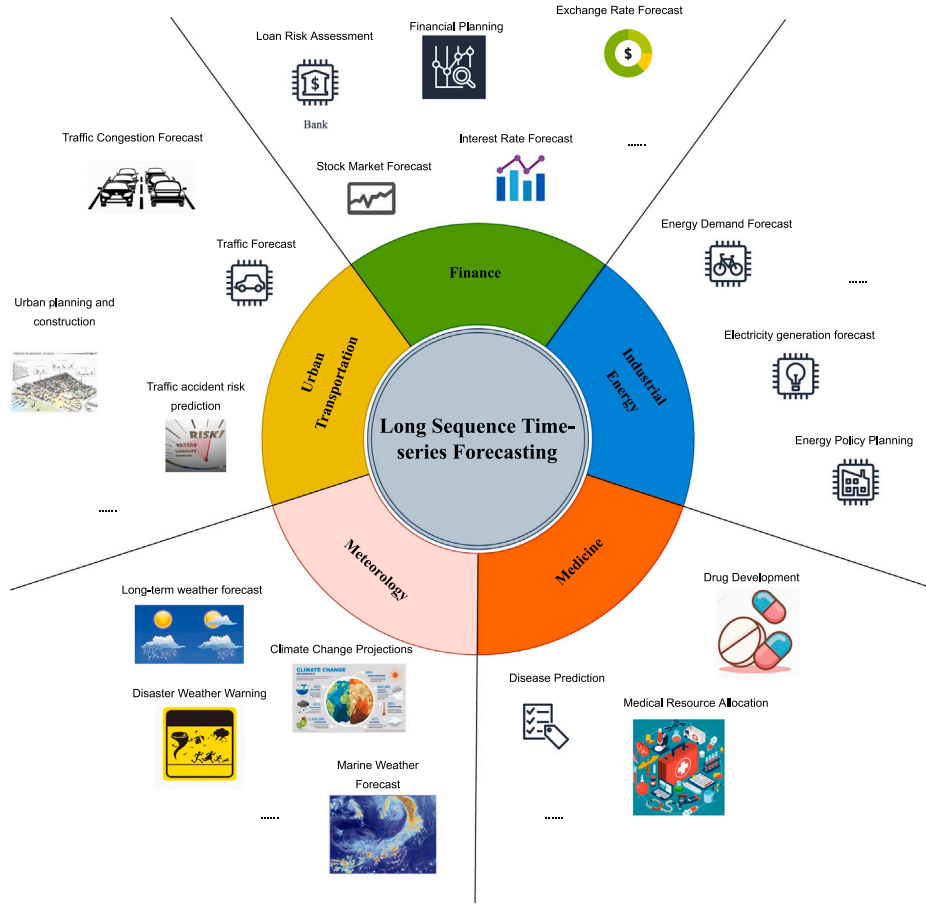


Fig. 14. Applications of LSTF.

structure. Recently, although LSTM, GRUs and other models combined with attention have exhibited better performance in terms of capturing long-term dependencies, they perform poorly on multivariate LSTF problems. DSANet [78] captures complex mixed global and local temporal patterns via two parallel convolutions and incorporates an attention mechanism to optimize the long-term correlation capture problem for multivariate LSTF.

The Transformer [32], on the other hand, is a model that combines various mechanisms, such as attention, embedding and encoder-decoder structures, for the field of NLP. Later, studies improved the Transformer and gradually applied it to TSF, imaging and other fields, gradually making Transformers their own genre. Therefore, this paper summarizes this class of models as the Transformer class. The performance of Transformers on LSTF is amazing, and they have gradually become the current mainstream approaches. The LogSparse Transformer [91] model first introduced a Transformer to the field of TSF and improved the prediction accuracy achieved for fine-grained, long-term dependent time series in a case with a limited memory budget. This study verified the effectiveness of Transformers in LSTF. Subsequently, much work regarding LSTF has revolved around the improvement exhibited by Transformers. When dealing with time-series data [92], the embedding mechanism of the vanilla Transformer often fails to fully encode long sequences and is prone to losing long-term dependencies. To address this problem, SpringNet [94] improves the model's ability to capture long-term repeatable fluctuating sequence correlations in the embedding postinformation by means of a DWT attention layer. TCCT [98] integrates CSPNet with a self-concentration mechanism and distills it with an expansive causal convolution layer to obtain more granular information with a negligible additional computational cost; it also improves the obtained information to enhance the LSTF capability

Table 10
Dataset details.

Dataset	Time Horizon and Data Granularity	Features
ETTh1	2016.07–2018.07 1 h	7 features
Stock	2000.01.01–2022.02.28 1 d	12 features
PEMS03	2018.01.09–2018.11.30 5 min	358 features
WTH	2010.01.01–2013.12.31 1 h	12 features
COVID19	2020.1.22–now 1d	4 features

of the model without increasing its computational cost. In contrast, Autoformer [33] uses an Auto-Correlation mechanism to break through the information utilization bottleneck of Transformers and achieve an LSTF accuracy improvement. AST [93] introduces the combination of a GAN and a Transformer to optimize the Transformer model in LSTF to address the problem of cumulative error generation.

4.3. Computational complexity and memory requirements

The length of a sequence can impede efficient data processing [33] in LSTF, as the computational complexity grows exponentially with the length of the sequence. Additionally, storing the entire sequence in memory can be challenging due to limited computer memory availability [96]. Numerous studies have been conducted to enhance the efficiency of LSTF. The time complexity of self-attention computation in a Transformer is initially established at $O(n^2)$, which inevitably leads to elevated computational expenses; subsequent works on LSTF involving Transformers have optimized this time complexity. Log Trans [91] reduces the time complexity to $O(L(\log L)^2)$ via convolutional self-attention, while Informer [17] obtains a time complexity of $O(L\log L)$

Table 11
Case model.

Model	Year	Published Journals/Conferences
Pyraformer	2022	ICLR
FEDformer	2022	ICML
Autoformer	2021	NeurIPS
Informer	2021	AAAI
Reformer	2020	ICLR
Transformer	2017	NeurIPS
MTGNN	2020	SIGKDD
Graph WaveNet	2019	IJCAI
LSTNet	2018	SIGIR

through ProbSparse self-attention achieves $O(L \log L)$ memory usage, and drastically reduces the spatial complexity of the model computations with the self-attention distillation operation. In contrast, Autoformer [33] achieves $O(L \log L)$ computational time complexity through sequence decomposition and the Auto-Correlation mechanism, which breaks the information utilization bottleneck of the vanilla Transformer. Pyraformer [96] achieves $O(L \log L)$ computational time complexity through pyramidal attention by letting the maximum length of the signal traversal path be a constant with respect to the sequence length L , achieving time and space complexity levels on the order of $O(L)$. FEDformer [97] achieves $O(L)$ linear computational complexity and storage overhead through frequency domain mapping. In other works, NET3 [114] effectively reduces the number of parameters in the model using tensor decomposition, and the STRN [132] significantly reduces the number of parameters required for the model through meta-learning and a unique framework design for CNNs and GCNs. Each work has its own unique perspective for optimizing the parametric part of the model, and a general architecture and component that can reduce the number of required model parameters has not yet emerged.

4.4. Performance evaluation method for LSTF

To address the third key problem regarding the LSTF evaluation methodology, we are not currently aware of any work in this area. Therefore, we apply the Kruskal–Wallis test to evaluate the performance of an LSTF model from two perspectives: model-model and model-self.

From the model-self perspective, in the LSTF task, we believe that when the prediction horizon of a model exceeds a certain range and the prediction accuracy decreases significantly, the larger the range of the prediction horizon is, the better the LSTF performance of the model. Therefore, how to define this range becomes an important problem. For different models, this value is different. Let us use the example in Fig. 12 to illustrate that for *model 1*, with an input time step size of 48 and a prediction horizon of 24, the prediction accuracy of this model starts to decrease significantly; then, for *model 1*, prediction horizons > 24 can be considered LSTF. In contrast, for *model 2*, the prediction accuracy only decreases significantly when the prediction horizon > 120 ; then, for *model 2*, the prediction horizon of 24 is obviously not its defined value for LSTF. Therefore, the LSTF performance of *model 2* is better.

From the model-model perspective, we consider that in the same LSTF task, the model with superior prediction accuracy between different models and statistically significantly better than another with the same predict horizon can be considered as more LSTF performance. For instance, in Fig. 13, when *model 1* and *model 2* are evaluated on the same LSTF task with identical input time steps and prediction horizons, *model 2* exhibits significantly higher prediction accuracy than *model 1*. Hence, we consider that *model 2* outperforms *model 1* in terms of LSTF performance under these conditions.

Furthermore, we believe that comparing the accuracies of models purely based on the numerical magnitudes of their evaluation metrics is not reliable, and we need to demonstrate that the prediction accuracy of one model is superior to that of another model in a statistical sense. Therefore, we propose to compare whether there is a significant difference between the data distributions of the differences between the true predicted values of different models using the Kruskal–Wallis test [152]. Compared with other common hypothesis testing methods such as standard t-tests and ANOVA, the Kruskal–Wallis test has the advantage of not assuming that the data follows a normal distribution [153], making it suitable for comparative analysis of non-normally distributed data. At the same time, it can detect differences of any form between different groups, without assuming that they have the same variance or the same distribution [154], making it widely applicable. However, the Kruskal–Wallis test also has its limitations: it is a non-parametric testing method, and therefore requires a larger sample size to achieve higher statistical power [155], which can be satisfied by the long sequence data in LSTF.

The Kruskal–Wallis test is applied to compare the variances of multiple continuous datasets when the conditions of the normal distribution are not satisfied. The idea of the Kruskal–Wallis test is to mix n groups of samples into one dataset (assuming that they are from the same sample) and then encode the data from small to large. Each data point has its own rank in the mixed dataset; if two orders are the same, the average value is taken as the rank. If the n groups are acquired from the same sample, then the rank of each group should not differ much from the total average rank of the mixed data. If the difference is large, then the groups are not from the same overall distribution; i.e., there is a significant difference between the data.

The Kruskal–Wallis test procedure is as follows.

- (1) Suppose that there are m mutually independent simple random samples $\{X_1, \dots, X_{n_i}\} (i = 1, \dots, m)$ and that all $N = \sum_{i=1}^m n_i$ observations of each sample are arranged in increasing order.
- (2) Denote by $R_i (i = 1, \dots, m)$ the sum of the ranks of n_i observations $\{X_1, \dots, X_{n_i}\}$ for the i th sample in this arrangement.
- (3) Calculate the statistic H :

$$H = \frac{12}{N(N+1)} \sum_{i=1}^m \frac{R_i^2}{n_i} - 3(N+1) \quad (12)$$

If each sample has r identical data and $t_i (i = 1, \dots, r)$ is the number of times that the i th common observation of each sample occurs in all N observations, then the following modified statistic H' is calculated (H and H' approximately follow a chi-square distribution when N is sufficiently large with degrees of freedom $v = m - 1$):

$$H' = \frac{N(N^2 - 1)}{\sum_{i=1}^r (t_i^3 - t_i)} H \quad (13)$$

- (4) For a given significance level α and $v = m - 1$ degrees of freedom, χ^2 is obtained as the upper lateral quantile of the distribution $\chi_{\alpha, v}^2$, and the hypothesis of no significant difference between the samples is rejected when $H \geq \chi_{\alpha, v}^2$ (or $H' \geq \chi_{\alpha, v}^2$), which means that the m samples are considered to be statistically significantly different from each other.

5. Applications and datasets

In this section, we summarized the applications of LSTF in the financial, industrial energy, urban transportation, medicine, and meteorological fields, as shown in Fig. 14, as well as the publicly available datasets in each application field. We collected commonly used datasets from various literature and presented the basic information of each dataset for reference. The survey results are presented in Appendix A.

Table 12

Multivariate experimental forecasting results obtained on ETTh1. The MAE and RMSE mark the top two models with the best prediction results under each horizon, where a lower RMSE or MAE indicates a better prediction effect.

Model	Metric	ETTh1				
		Horizons				
		12	48	192	384	768
Informer	MAE	0.474	0.631	0.809	0.884	0.878
	RMSE	0.676	0.829	1.010	1.090	1.084
	MAPE	10.501	13.583	12.135	13.221	14.752
Autoformer	MAE	0.422	0.444	0.479	0.489	0.519
	RMSE	0.616	0.654	0.706	0.711	0.726
	MAPE	11.158	11.272	11.998	12.425	14.027
Reformer	MAE	0.532	0.613	0.716	0.742	0.825
	RMSE	0.733	0.839	0.956	0.983	1.084
	MAPE	12.300	12.081	8.601	13.077	23.151
Pyraformer	MAE	0.444	0.547	0.676	0.747	0.780
	RMSE	0.630	0.743	0.881	0.950	0.978
	MAPE	10.188	9.166	10.972	13.801	23.171
FEDformer	MAE	0.368	0.390	0.444	0.466	0.506
	RMSE	0.541	0.581	0.649	0.675	0.708
	MAPE	9.709	9.694	11.738	12.512	13.878
Transformer	MAE	0.485	0.668	0.779	0.845	0.817
	RMSE	0.68	0.863	0.979	1.051	1.019
	MAPE	10.908	10.629	9.767	13.96	20.209
MTGNN	MAE	0.35	0.413	0.515	0.661	0.694
	RMSE	0.531	0.609	0.722	0.863	0.895
	MAPE	8.18	8.508	8.776	11.825	14.032
LSTNet	MAE	0.609	0.631	0.71	0.781	0.829
	RMSE	0.827	0.852	0.931	0.998	1.031
	MAPE	10.237	10.23	11.006	14.646	20.746
Graph WaveNet	MAE	0.401	0.432	0.499	0.533	0.581
	RMSE	0.61	0.653	0.728	0.759	0.793
	MAPE	8.775	8.29	8.556	11.146	14.119

5.1. Finance

In the finance, LSTF is often used for predicting economic cycles [51], fiscal cycles, and long-term trends in stocks [157], and can assist financial traders in making more favorable decisions or help companies make better financial decisions based on predicted revenue and expenditures. For example, in the stock market, LSTF can assist investors in developing more accurate investment strategies, predicting trends and fluctuations in future stock prices [1], to make better investment decisions. In financial planning, LSTF can assist individuals and businesses in financial planning [26], predicting future financial conditions such as income, expenses, and profitability, in order to better plan financial goals and fund operations. In addition, LSTF can also assist financial institutions in loan risk assessment, predicting future borrower repayment ability and credit risk [158], or predicting future interest rate trends to better formulate monetary and interest rate policies. We summarized the prediction tasks and open-source datasets in the finance field in recent years in [Appendix A.1](#).

5.2. Industrial energy

In the industrial energy field, LSTF is often used to assist in formulating long-term strategic resource planning strategies [25]. As shown in [Appendix A.2](#), there are many works and open-source datasets related to industrial energy prediction. LSTF can help companies and governments predict future energy demands, such as electricity, oil, natural gas, etc., in order to better plan energy production and supply.

It can also help power companies predict future power generation [54] to ensure adequate and stable supply. In addition, LSTF can also help governments and companies develop energy policy planning or manage energy supply chains [159]. Long-term forecasting in the industrial energy field has applications in many aspects, which can help companies and governments better plan and manage, reduce risks, improve efficiency, and achieve sustainable development.

5.3. Urban transportation

In the field of urban transportation, LSTF has been applied in various aspects. It can assist urban transportation management departments in predicting future traffic flow [160] for better traffic planning and management. LSTF can also be used to forecast future traffic congestion, which can help in formulating strategies and measures to alleviate congestion [2]. In addition, long-term forecasting can assist urban transportation management departments in predicting future traffic accident risks and traffic safety issues [161], enabling better traffic safety management and accident prevention. The datasets are shown in [Appendix A.3](#).

5.4. Meteorology

The application of LSTF in meteorology is generally focused on long-term trend forecasting of climate, providing decision support for industries such as agriculture, marine transportation, and natural weather disaster warning. For example, LSTF can be used to forecast long-term

Table 13

Multivariate experimental forecasting results obtained on Stock and COVID19. The MAE, RMSE, and MAPE mark the top two models with the best prediction results under each horizon, and a lower RMSE or MAE indicates a better prediction effect.

Model	Metric	Stock					COVID19				
		Horizons					Horizons				
		12	48	96	288	480	12	24	48	60	72
Informer	MAE	9.695	10.193	10.268	9.839	9.897	1.556	1.591	1.548	1.534	1.573
	RMSE	12.653	13.171	13.260	12.738	12.582	1.805	1.891	1.789	1.746	1.803
	MAPE	0.824	0.823	0.859	0.856	0.928	0.957	0.974	1.024	1.047	1.046
Autoformer	MAE	1.041	1.335	1.636	2.893	3.664	0.176	0.212	0.269	0.170	0.205
	RMSE	1.567	2.006	2.510	4.375	5.517	0.274	0.355	0.418	0.241	0.316
	MAPE	0.249	0.285	0.315	0.419	0.450	0.161	0.185	0.221	0.161	0.198
Reformer	MAE	9.319	9.704	9.744	9.917	9.745	1.579	1.693	1.608	1.533	1.574
	RMSE	12.209	12.606	12.654	12.721	12.437	1.855	2.001	1.878	1.769	1.824
	MAPE	0.753	0.853	0.924	0.929	0.958	0.939	0.972	0.933	0.948	0.942
Pyraformer	MAE	9.662	10.015	10.033	10.183	10.194	1.747	1.781	1.756	1.702	1.713
	RMSE	12.581	13.004	13.074	13.073	12.938	2.054	2.080	2.032	1.981	1.984
	MAPE	0.866	0.922	0.929	0.944	0.980	1.155	1.214	1.201	1.133	1.181
FEDformer	MAE	1.082	1.350	1.649	2.899	3.683	0.161	0.195	0.256	0.174	0.218
	RMSE	1.593	2.019	2.523	4.381	5.531	0.269	0.339	0.406	0.243	0.325
	MAPE	0.291	0.303	0.327	0.424	0.461	0.129	0.160	0.206	0.164	0.197
Transformer	MAE	9.394	9.611	9.564	9.350	9.563	1.586	1.689	1.634	1.609	1.657
	RMSE	12.28	12.501	12.501	12.292	12.13	1.806	1.944	1.909	1.845	1.909
	MAPE	0.765	0.766	0.780	0.788	0.866	1.085	1.143	0.977	1.245	1.208
MTGNN	MAE	8.664	8.953	9.359	9.483	9.704	0.994	1.167	1.488	1.501	1.586
	RMSE	11.391	11.758	12.175	12.272	12.398	1.340	1.559	1.924	1.886	1.989
	MAPE	0.713	0.751	0.807	0.822	0.889	0.431	0.548	0.713	0.759	0.802
LSTNet	MAE	7.734	7.917	8.001	8.404	8.393	2.098	2.122	1.971	1.939	2.081
	RMSE	10.552	10.728	10.837	11.162	11.193	2.658	2.636	2.397	2.389	2.551
	MAPE	1.034	1.11	1.211	1.167	1.117	1.322	1.528	1.334	1.268	1.399
Graph WaveNet	MAE	10.011	10.377	10.464	10.605	10.646	0.641	0.778	0.873	0.879	1.002
	RMSE	13.424	13.650	13.602	13.547	13.385	1.008	1.195	1.255	1.203	1.365
	MAPE	0.739	0.795	0.849	0.904	0.939	0.216	0.267	0.321	0.351	0.406

climate change [162], including global climate change trends, temperature, precipitation, and other indicators, providing scientific basis for national policy-making in response to climate change. LSTF can also be used to issue early warnings for natural weather disasters such as heavy rain, floods, and typhoons [3], allowing for adequate preparation and mitigation of potential harm to human life and property. Additionally, LSTF can predict information such as sea surface temperature and marine meteorology for the next few months or years [163], providing decision support for industries such as fishing and marine transportation. The datasets are shown in Appendix A.4.

5.5. Medicine

The development of the medical field bears the responsibility of safeguarding and advancing human society. LSTF can be applied to various stages of drug development. For instance, it can predict the toxicity, pharmacokinetics, pharmacodynamics, and other parameters of a drug, helping researchers optimize its design and screening process [164]. Additionally, long-term forecasting can be used to predict the market prospects and sales of a particular drug, aiding in the development of marketing strategies. LSTF can also predict future medical needs over a certain period of time, such as the population structure and incidence of chronic diseases in a particular region [165]. These forecasting results can be utilized for the rational allocation and planning of medical resources. The datasets are shown in Appendix A.5.

6. Performance evaluation

In this section, we discuss prediction performance evaluation metrics in the field of TSF based on nearly 80 pieces of literature. Here, we employ the classification method introduced by [166] to divide the prediction accuracy metrics into three groups: scale-dependent, scale-independent, and scaled error metrics, which are based on whether the evaluation metrics are affected by the data scale and the way in which the data scale effects are eliminated. The evaluation methods used in these papers are summarized and described in detail in Table 9. Moreover, we analyze the characteristics of each metric appearing in the papers in terms of the above classification metrics.

6.1. Scale-dependent measures

Scale-dependent measures are evaluation metrics whose data scales depend on the data size of the original data, and they are the most widely used evaluation metrics in forecasting. The advantage of this type of metric is that it is computationally simple and can be used to measure different methods applied to the same dataset for comparison purposes. Notably, the limitation of this type of metric in TSF is that it is not possible to compare algorithm models across datasets with different scales, and such an evaluation metric is affected by datasets with particularly large data scales, which is not conducive to subsequent comparative studies. We summarize the current popular scale-dependent measures as follows.

$$\bullet \text{ MAE} = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$$

Table 14

Multivariate experimental forecasting results obtained on PEMS03 and WTH. The MAE, RMSE, and MAPE mark the top two models with the best prediction results under each horizon, and a lower RMSE or MAE indicates a better prediction effect.

Model	Metric	PEMS03					WTH				
		Horizons					Horizons				
		12	48	192	384	768	12	48	192	384	768
Informer	MAE	0.211	0.246	0.269	0.261	0.283	0.317	0.427	0.551	0.592	0.584
	RMSE	0.327	0.373	0.405	0.400	0.449	0.517	0.622	0.763	0.799	0.787
	MAPE	2.903	3.452	4.048	3.914	4.355	1.086	1.491	1.914	2.108	2.002
Autoformer	MAE	0.282	0.409	0.615	0.676	0.673	0.392	0.494	0.566	0.581	0.582
	RMSE	0.389	0.549	0.796	0.885	0.869	0.583	0.699	0.774	0.791	0.797
	MAPE	3.119	3.906	5.488	6.867	6.309	1.368	1.715	1.968	2.085	2.122
Reformer	MAE	0.208	0.261	0.269	0.271	0.275	0.303	0.411	0.493	0.507	0.505
	RMSE	0.324	0.405	0.431	0.437	0.447	0.503	0.611	0.695	0.709	0.705
	MAPE	2.754	4.047	4.188	4.212	4.333	1.03	1.382	1.644	1.717	1.728
Pyraformer	MAE	0.212	0.243	0.273	0.274	0.291	0.304	0.406	0.511	0.529	0.530
	RMSE	0.329	0.371	0.408	0.412	0.443	0.501	0.602	0.712	0.731	0.727
	MAPE	2.872	3.495	4.081	4.096	4.628	1.023	1.389	1.776	1.946	1.931
FEDformer	MAE	0.229	0.269	0.332	0.35	0.364	0.345	0.434	0.539	0.563	0.572
	RMSE	0.328	0.384	0.462	0.496	0.517	0.533	0.632	0.752	0.779	0.786
	MAPE	0.893	3.376	4.401	4.487	4.809	1.184	1.519	1.868	1.979	2.019
Transformer	MAE	0.199	0.241	0.258	0.253	0.265	0.321	0.435	0.567	0.553	0.547
	RMSE	0.313	0.369	0.391	0.387	0.421	0.514	0.637	0.741	0.762	0.749
	MAPE	2.684	3.403	3.949	3.804	3.936	1.087	1.557	1.883	2.097	1.991
MTGNN	MAE	0.163	0.218	0.285	0.316	0.356	0.304	0.404	0.505	0.526	0.562
	RMSE	0.249	0.333	0.418	0.459	0.513	0.508	0.606	0.703	0.723	0.756
	MAPE	2.309	3.026	4.536	4.681	5.260	1.001	1.384	1.776	1.932	2.121
LSTNet	MAE	0.259	0.302	0.360	0.362	0.380	0.361	0.439	0.522	0.544	0.569
	RMSE	0.383	0.431	0.499	0.507	0.533	0.547	0.630	0.717	0.738	0.763
	MAPE	3.787	4.479	5.258	5.189	5.479	1.211	1.489	1.786	1.937	1.997
Graph WaveNet	MAE	0.198	0.334	0.449	0.426	0.448	0.336	0.456	0.546	0.571	0.608
	RMSE	0.295	0.486	0.619	0.597	0.62	0.545	0.661	0.749	0.769	0.804
	MAPE	2.679	4.256	5.862	5.472	5.816	1.074	1.483	1.782	1.976	1.966

Table 15

Kruskal–Wallis test results. $P < 0.05$ denotes significant difference; Statistic denotes measure the degree of variation between samples, larger values indicate greater variation between samples.

Model comparison	ETTh1		Stock	
	Statistic	P-value	Statistic	P-value
Autoformer-Informer	445.97	0.013	493.54	0.002
Autoformer-FEDformer	99.85	0.083	5.24	0.611
Autoformer-Pyraformer	446.48	0.003	295.78	0.023
Autoformer-Transformer	341.68	0.023	516.95	0.001
Autoformer-Reformer	361.18	0.044	494.34	0.009
Autoformer-LSTnet	402.81	0.034	311.62	0.012

- $MSE = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$
- $LAE = \max\{y_i - \hat{y}_i, i = 1, \dots, m\}$
- $RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$
- $MSLE = \frac{1}{m} \sum_{i=0}^m (\log(y_i + 1) - \log(\hat{y}_i + 1))^2$
- $MALE = \frac{1}{m} \sum_{i=1}^m |\log y_i - \log \hat{y}_i|$
- $RM SLE = \sqrt{\frac{1}{m} \sum_{i=1}^m (\log y_i - \log \hat{y}_i)^2}$

Here $\hat{y}_i$ denotes the predicted value, y_i denotes the true value, m denotes the sample size, and $\bar{y}_i$ denotes the mean of the true sample data.

Among all scale-dependent measures, the root mean square error (RMSE) and mean absolute error (MAE) are almost the most widely

used metrics. The LAE metric can only reflect the maximum error situation in the prediction results, and is susceptible to outliers and cannot reflect the true fit of the results, so it is less used in DL in recent years. The RMSE is effective due to its simplicity and close relationship with statistical modeling. For unbiased forecasting methods, it gives exactly the same value as the variance of the forecast error. The main difference between the RMSE and MSLE is their sensitivity to outliers in the given dataset. For MSLE, the effect of outliers can be reduced, while the RMSE is relatively more sensitive to outlier data points. The MSLE is more robust when evaluating datasets containing outliers. In contrast, the MAE better reflects the true error situation than the RMSE. The RM SLE is useful in situations where there are significantly more penalties for underpredicted valuations than for overpredicted valuations.

Table A.16

Finance application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
Gold prices	[125]	2014.1–2018.4	Day	The dataset contains daily gold prices (U.S. dollars) from January 2014 to April 2018 from http://finance.yahoo.com and includes minimum, mean, maximum, median, standard deviation (SD), skewness and kurtosis used to describe the nature of the distribution.
GECom2014 Electricity Price	[150]	–	Hour	The dataset was published in the GECom 2014 Forecasting Competition, which consists of four questions: electricity load forecasting, electricity price forecasting, and two additional questions related to wind and solar power generation. In the electricity price forecasting, the dataset has 21552 time points with hourly data granularity. Data source: https://www.dropbox.com/s/pqenrr2mcvl0hk9/GECom2014.zip?dl=0 .
Exchange-Rate	[33,82,104,143]	1990–2016	Day	The dataset collects daily exchange rates from 1990 to 2016 for eight countries, including Australia, the United Kingdom, Canada, Switzerland, China, Japan, New Zealand and Singapore. Data source: https://github.com/laiguokun/multivariate-time-series-data .
S&P 500	[126]	2001.1–2017.5	Day	This dataset records the daily S&P 500 index from 2001.01–2017.05, with a total of 4506 data. Data source: http://finance.yahoo.com .
Shanghai Composite	[126]	2005.1–2017.6	Day	This dataset records the daily SSE indices from 2005.01–2017.06, with 2550 data. Data source:.
S&P500 Stocks	[71]	2013.2.8–2018.2.7	Day	The dataset consists of 505 common stocks issued by 500 large-cap companies and traded on the American Stock Exchange, recording historical daily stock prices from 2013.2.8–2018.2.7 for all companies currently included in the S&P 500 index. Data source: https://www.kaggle.com/camnugent/sandp500 .
CRSP's Stocks	[151]	–	Day	The dataset is from CRSP and includes data on individual stock returns and prices, S&P 500 index returns, industry categories, number of shares outstanding, ticker symbols, exchange codes and trading volume. Data source: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html .
Gas Station Revenue	[78]	2015.12.1–2018.12.1	Day	This dataset records the daily revenue of five gas stations, which are geographically close to each other and have interrelated effects, for the period from December 1, 2015 to December 1, 2018. Data source: https://github.com/bighuang624/DSANet/tree/master/data .
Finance Japan	[72]	2003.1–2016.12	4 months	The dataset was collected by the Ministry of Finance of Japan and records general partnerships, limited partnerships, limited liability companies and joint stock companies from the first quarter of 2003 to the fourth quarter of 2016. The time granularity is available on an annual and quarterly basis. The total number of companies surveyed for the quarter is 57,775 and the total number of companies surveyed for the year is 60,516. Data source: https://www.mof.go.jp/english/pri/reference/ssc/outline.htm .
Stock Opening Prices	[106]	2007–2016	Day	The dataset collects daily opening prices for 50 stocks in 10 sectors in Financial Yahoo from 2007–2016. Each sector involves 5 top companies and each stock contains 2518 data. Data source: https://github.com/z331565360/State-Frequency-Memory-stock-prediction .

6.2. Scale-independent measures

Scale-independent measures, in contrast to scale-dependent measures, are evaluation metrics that are not affected by the size of the

original data. These evaluation metrics also have some drawbacks: some evaluation metrics do not consider the case when the actual value is zero (MAPE), and some evaluation metrics have an incorrectly

Table A.17
Industrial energy application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
Power Consumption	[106]	2006.12–2010.11	Minute	This dataset records the electricity consumption of a household over a period of nearly 4 years, including voltage, electricity consumption and other characteristics, at a data granularity of min. Data source: https://archive.ics.uci.edu/ml/datasets/Individual+household+electric+power+consumption .
Solar-Energy	[82,91,93,104,143]	2006	5 min	This dataset records the highest solar power production from 137 photovoltaic plants in Alabama in 2006, with a data granularity of 5-minute sampling. Data source: https://www.nrel.gov/grid/solar-power-data.html .
Electricity	[33,82,93,104,143]	2011–2014	15 min	This dataset records the electricity consumption of 321 customers from 2011 to 2014, with electricity consumption measured in kWh and data granularity of 15 min. Data source: https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014 .
Wind	[91,93]	1986–2015	Hour	This dataset records hourly estimates of energy potential as a percentage of the maximum output of power plants for a European region for the period 1986–2015. Data source: https://www.kaggle.com/sohier/30-years-of-european-wind-generation .
ETT	[17,33]	2016.7–2018.7	15 min	This dataset records the load and oil temperature of power transformers recorded every 15 min from July 2016 to July 2018. Has ETTh1, ETTh2 at hourly granularity and ETTm1 at 15 min granularity. Data source: https://github.com/zhouhaoyi/ETDataset .
Sanyo	[94]	2011.1.1–2017.12.31	Day	This dataset records solar power generation data for two photovoltaic power plants in Alice Springs, Northern Territory, Australia, including solar data from January 1, 2011 to December 31, 2017. Data source: http://dkasolarcentre.com.au/source/alice-springs/dka-m4-b-phase .
Hanergy	[94]	2011.1.1–2016.12.31	Day	This dataset records solar power generation data from two photovoltaic plants in Alice Springs, Northern Territory, Australia, and contains data from January 1, 2011 to December 31, 2016. Data source: https://dkasolarcentre.com.au/source/alice-springs/dka-m16-b-phase .
SGSMEPC	[141]	2014.1.1–2015.2.19	Day	The dataset records grid data from 2014.01.01–2015.02.19 of State Grid Shanghai Electric Power Company with a data granularity of 1 day.

skewed distribution when the actual value is close to zero. We collect and list the most popular scale-independent measures:

- $MAPE = \frac{100}{m} \sum_{i=1}^m \left| \frac{y_i - \hat{y}_i}{y_i} \right|$
- $IA = \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (|y_i - \hat{y}_i| + |y_i - \hat{y}_i|)^2}$
- $RRSE = \frac{\sqrt{\sum_{i=1}^m (y_i - \hat{y}_i)^2}}{\sqrt{\sum_{i=1}^m (y_i - \hat{y}_i)^2}}$
- $R^2 = 1 - \frac{\sum_{i=1}^m (y_i - \hat{y}_i)^2}{\sum_{i=1}^m (y_i - \bar{y})^2}$
- $sMAPE = \frac{1}{m} \sum_{i=1}^m \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}$
- $NRMSSE = \frac{\sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}}{y_{\max} - y_{\min}}$, where $y_{\max}$ is the maximum value of the real sample data and $y_{\min}$ is the minimum value of the real sample data;
- $RAE = \frac{\sum_{i=1}^m |y_i - \hat{y}_i|}{\sum_{i=1}^m |y_i - \bar{y}|}$
- $RMSPSE = \sqrt{\frac{1}{m} \sum_{i=1}^m \left(\frac{y_i - \hat{y}_i}{y_i} \right)^2}$

- $NMAE = \frac{\sum_{i=1}^m |y_i - \hat{y}_i|}{\sum_{i=1}^m y_i}$

By scaling the error according to the data scale of the real data, the value of a scale-independent measure may be limited to an interval range, such as [0, 1]. Therefore, the use of scale-independent measures can largely avoid the influence of the scale of the real data, which has an impact on the value of the evaluation metric. In addition, we note that such an evaluation metric has some defects due to the presence of scale factors: (1) first, for the case where the true value is 0, some evaluation metrics have undefined situations involving a denominator of 0, such as the MAPE; (2) when the true value is close to 0, some evaluation metrics produce skewed distributions, such as the MAPE, and Makridakis [167] argued that the skewed distribution of the MAPE is due to the penalty inequality between positive and negative prediction errors. Therefore, they proposed the sMAPE to solve this problem, while Goodwin and Lawton [168] pointed out that the sMAPE is still asymmetric.

Table A.18
Urban transportation application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
Paris metro line	[27]	2009–2010	Hour	This dataset records the passenger flow on Paris metro lines 3 and 13 and consists of 1742 observations with a data granularity of 1 hour. The dataset is derived from the Artificial neural network and computational intelligence forecasting competition http://www.neural-forecasting-competition.com/ .
PeMS03, PeMS04, PeMS07, PeMS08	[33,82–84,93,104,131,143]	–	30 s	The data comes from the PeMS system. The system records traffic flow data for areas such as the Bay Area in California, where traffic data is collected every 30 s and aggregated over a five-minute period. Each sensor on the freeway has traffic flow, occupancy and speed for each time stamp. The sensors are inductive loop traffic detector devices on the mainline, exit ramp or entrance ramp locations. Users can download the data for the desired time period on demand. This data was obtained from http://pems.dot.ca.gov/ .
Birmingham Parking	[71]	2016.10.4–2016.12.19	30 min	This dataset records the parking lot ID, parking lot capacity, parking lot occupancy and update time for 30 parking lots operated by Birmingham National Car Park. The data period is 8:00 16:30 per day from 2016.10.4–2016.12.19, and the data granularity is 30 min. Data source: http://archive.ics.uci.edu/ml/datasets/parking+birmingham .
METR-LA	[82,83]	2012.3.1–2012.6.30	5 min	This dataset records traffic information collected by loop detectors on Los Angeles County freeways. Data source: https://drive.google.com/drive/folders/10FOTa6HXPqX8Pf5WRoRwCfNw9BrNZEIX .
PEMS-BAY	[82,83]	2017.1.1–2017.5.13	30 s	This dataset records traffic speed readings from 325 sensors collected by PeMS, the California Transit Agency Performance Measurement System. Data source: https://drive.google.com/drive/folders/10FOTa6HXPqX8Pf5WRoRwCfNw9BrNZEIX .
SPMD	[36]	2015.5.10–2015.10.18	Hour	The dataset was extracted by the University of Michigan Transportation Research Institute and recorded the driving records of approximately 3,000 drivers in Ann Arbor, Michigan between May 10, 2015 and October 18, 2015. Data source: https://github.com/ElmiSay/DeepFEC .
VED	[36]	2017.11–2018.11	Hour	This dataset contains the fuel and energy consumption of various personal vehicles (cars, convertibles, pickup trucks, SUVs, etc.) operating under different realistic driving conditions in Michigan, USA from November 2017 to November 2018. Data source: https://github.com/ElmiSay/DeepFEC .
England	[84]	2014.1–2014.6	15 min	The dataset is derived from UK freeway traffic data made public by the UK government, which was selected for the period January 2014 to June 2014, and the data contains national average speeds and traffic volumes. Data source: http://tris.highwaysengland.co.uk/detail/trafficflowdata .
TaxiBJ+	[132]	–	30 min	This dataset records the distribution and trajectory of more than 3000 cabs in Beijing. The data granularity is 30 min. time span is, respectively, P1: 2013.07.01–2013.10.31; P2: 2014.02.01–2014.06.30; P3: 2015.03.01–2015.06.30; P4: 2015.11.01–2016.03.31.
HappyValley	[132]	2018.1.1–2018.10.31	Hour	This dataset records the hourly population density of popular theme parks in Beijing from 01.01.2018 to 31.10.2018.
NYC Taxi	[111]	2009.1.1–2016.6.30	Hour	This dataset records details of every cab trip in New York City from 2009.01.01 to 2016.06.30.

Table A.19
Meteorological application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
Weather pollutants	[109]	2015.1.1–2016.4.1	Hour	The dataset contains data of pollutants (PM2.5, PM10, NO2, SO2, O3, and CO) sampled hourly for 76 stations in the Beijing–Tianjin–Hebei region from January 1, 2015 to April 1, 2016 (10944 h in total). Hour-by-hour meteorological observations, including wind speed, wind direction, temperature, barometric pressure, and relative humidity, for the same period were also downloaded from the China Weather website platform, which is maturely maintained by the China Meteorological Administration (CMA).
Beijing PM2.5	[104]	2010.1.1–2014.12.31	Hour	The dataset contains hourly PM2.5 data and associated meteorological data for Beijing, China. PM2.5 measurements are the target series. The data include dew point, temperature, barometric pressure, combined wind direction, cumulative wind speed, hours of snowfall, and hours of rainfall for a total of 43,824 multivariate series. The dataset was obtained from https://archive.ics.uci.edu/ml/datasets.html .
Hangzhou Temperature	[126]	2011.1–2017.1	Day	This dataset records the daily average temperature of Hangzhou from 2011.01 to 2017.01, with a total of 2820 pieces of data. Data source: http://data.cma.cn/data/ .
WTH	[33]	2020	10 min	The dataset records weather conditions throughout 2020, with data recorded every 10 min, and contains 21 meteorological indicators such as temperature and humidity. Data source: https://www.bgc-jena.mpg.de/wetter/ .
USHCN	[11]	1887–2019	Day	The dataset records continuous daily meteorological records from 1887 to 2009 for 1219 stations collected from each state. The data include features such as average temperature, total daily precipitation, and other characteristics at a data granularity of days. Data source: https://www.ncdc.noaa.gov/ushcn/introduction .
KDD-CUP	[11]	2017.1–2017.12	Hour	This dataset is derived from the 2018 KDD CUP Challenge air quality dataset, which recorded PM2.5 measurements from 35 monitoring stations in Beijing from January 2017 to December 2017. Data source: https://www.kdd.org/kdd2018/kdd-cup .
US	[131]	2012–2017	Hour	This dataset records weather datasets from 2012 to 2017 from 36 weather stations in the U.S. The data granularity is hourly and the data contains features such as temperature, humidity, pressure, wind direction, wind speed, and weather description. Data source: https://www.kaggle.com/selfishgene/historical-hourly-weather-data .

6.3. Scaled errors

Scaled errors were first proposed by Hyndman and Koehler [166] and can be used to eliminate the effect of data size by comparing the prediction results obtained with the underlying method (usually the native method). The following scaled error is commonly used:

$$\begin{aligned} \bullet \text{ } MASE &= \text{mean} \left(\frac{|y_i - \hat{y}_i|}{\frac{1}{m} \sum_{i=2}^m |y_i - y_{i-1}|} \right) \\ \bullet \text{ } MdASE &= \text{median} \left(\frac{|y_i - \hat{y}_i|}{\frac{1}{m} \sum_{i=2}^m |y_i - y_{i-1}|} \right) \end{aligned}$$

There is a simple way to understand this evaluation metric; the denominator of this evaluation metric can be understood as the average error of the native predictions made one step ahead for the future. If the $MASE > 1$, then the experimental method under evaluation is worse than the native prediction, and vice versa. While MdASE is similar to MASE, the way MASE is calculated using the mean makes it more

susceptible to outliers, while the way MdASE is calculated using the median has stronger robustness and validity. However, such metrics can only reflect the results of comparison with the basic method and cannot visualize the error of the prediction results.

As seen from Table 9, the MAE, RMSE and MAPE are the most commonly used evaluation metrics. Among them, the MAE reflects the true error situation, while the RMSE magnifies the gaps between larger errors. The MAPE can be used to compare forecasts at different data scales and is easy to calculate. Therefore, the combination of the MAE, RMSE and MAPE can reflect the overall fitting effect, such as the real error situation and the outlier situation. Therefore, the MAE, RMSE and MAPE are used as evaluation indicators in the subsequent section.

7. Case study

In this section, we select the algorithm models of 10 excellent works based on the abovementioned works, and in Section 5 of this paper,

Table A.20
Medicine application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
ILI	[33,92]	2002–2021	Week	This dataset records data on patients with influenza-like illness (ILI) recorded weekly by the Centers for Disease Control and Prevention from 2002 to 2021, which describes the ratio of ILI patients to the total number of patients. Data source: https://gis.cdc.gov/grasp/fluview/fluportaldashboard.html .
COVID-19	[128]	2020.1.22–2020.6.17	Day	This dataset records daily data on confirmed and recovered cases collected from January 22, 2020–June 17, 2020 in six countries: Italy, Spain, Italy, China, the United States, and Australia. Data source: https://github.com/CSSEGISandData/COVID-19 accessed on 17/06/2020.
2020 OhioT1DM	[156]	2020	5 min	This dataset is the glycemic prediction dataset for the 2nd BGLP Challenge 2020. The dataset records 8 weeks of continuous glucose monitoring, insulin, physiological sensor, and self-reported life event data for each of 12 patients with type 1 diabetes. The specific dataset contains CGM glucose levels every 5 min; glucose levels from regular self-monitoring of blood glucose (finger sticks); push and basal insulin doses; self-reported meal times and carbohydrate estimates; self-reported exercise, sleep, work, stress, and illness times; and data from Basis Peak or Empatica Embrace bands. Data source: http://smarthealth.cs.ohio.edu/OhioT1DM-dataset.html .
MIMIC-III	[11]	2001–2012	Hour	The dataset is a public clinical dataset with over 58,000 admission records. It includes several clinical characteristics such as ICU stay data, glucose and heart rate.

we select five well-known benchmark datasets in the fields of finance, energy, meteorology, transportation and medicine for experiments and use the MAE, RMSE and MAPE as evaluation metrics based on the evaluation metric results presented in Section 6. Finally, we give the performance comparison for LSTF.

The selected datasets are as follows.

- **ETT**⁴: The ETT dataset is a long-term deployment dataset for electricity proposed by [82], which has several datasets with different temporal granularities, i.e., ETTh1, ETTh2 and ETTm1. The ETTh1 dataset is used in this paper, and it includes load and oil temperature values recorded every 1 hour during the period from July 2016 to July 2018.
- **Stock**⁵: The Stock dataset from <http://finance.yahoo.com> contains multiple features, such as the opening price, high price, low price, closing price, and trading volume, for each day from January 1, 2000, to February 28, 2022.
- **PEMS03**⁶: This dataset includes traffic data from the California Highway Network PEMS, a database collected by Caltrans on California freeways, as well as datasets from other Caltrans and partner agencies. We select 358 nodes in the San Francisco Bay Area freeways with sensor-measured road occupancy values collected every 5 minutes from January 9, 2018, to November 30, 2018.
- **WTH**⁷: This dataset is derived from weather data collected from more than 1,600 areas in the United States from 2010 to 2013. The data were collected every hour and contain 12 weather characteristics, including local humidity, for each area.

- **COVID19**⁸: This dataset contains confirmed coronavirus disease (COVID-19) cases, reported deaths, and reported recoveries for each day from 2020 to the present. The dataset includes time series data tracking the number of people affected by COVID-19 globally, documenting the numbers of confirmed cases of coronavirus infection that have been reported, deaths while infected with coronavirus, and recoveries reported from infections. The dataset is maintained by a team at the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University who collate data from around the world and update it daily.

The data granularities and feature information of these five datasets are shown in Table 10. Our implementation details are as follows.

We select six models based on the excellent work of the latest top-level conference on Transformers based on the work done in Section 3 of this paper: Pyformer [96], FEDformer [97], Informer [17], Autoformer [33], Reformer [172], a Transformer [32] and three other excellent models, the MTGNN [82], Graph WaveNet [173], and LST-Net, [105], as shown in Table 11.

We adopt the MSE as the loss function and use the adaptive moment estimation (ADAM) optimizer, where the initial learning rate is 10⁻⁴, the batch size is set to 32, the early stopping mechanism is used as the termination condition of the model training process, and the patience is set to 3. The Z-score normalization method is used for all datasets. The input_len is set to 96 for the ETTh1, PEMS03, WTH, and Stock datasets and 36 for the COVID19 dataset. The settings of predict_horizon are {12, 48, 192, 384, 768} for the ETTh1, PEMS03, and WTH datasets, {12, 48, 96, 288, 480} for the Stock dataset, and {12, 24, 48, 60, 72} for the COVID19 dataset. The training set, validation set, and test set of each datasets are divided according to a ratio of 7:1:2. The hyperparameter settings of each model are determined based on the

⁴ ETT: <https://github.com/zhouhaoyi/ETDataset>

⁵ Stock: <https://finance.yahoo.com/>

⁶ PEMS03: <https://github.com/Davidham3/STSGCN>

⁷ WTH: <https://www.bgc-jena.mpg.de/wetter/>

⁸ COVID19: <https://github.com/datasets/covid-19>

Table A.21
Competition application and dataset.

Dataset	Ref	Time range	Min-Granularity	Information
M5	[169]	–	Hour	The M5 contest is the latest in the M contest, which runs from March 2 to June 30, 2020. The contest dataset uses stratified sales data generously provided by Walmart, starting at the item level and then aggregated to departments, product categories and stores in three geographic regions of the U.S. (California, Texas and Wisconsin). In addition to time-series data, it includes explanatory variables that affect prices, such as price, promotion, day of the week and special events, which are used to improve forecast accuracy. Data source: https://github.com/Mcompetitions/M5-methods .
M4	[91,170]	–	Hour	The M4 dataset was created by randomly selecting 100,000 time series from the ForeDeCk database. the M4 dataset consists of time series of annual, quarterly, monthly and other (weekly, daily and hourly) data, which are divided into a training set and a test set. Data source: https://github.com/Mcompetitions/M4-methods/tree/master/Dataset .
M3	[171]	–	Hour	The M3 competition dataset contains 3003 time series data containing multiple series of Micro, Industry, Macro, Finance, Demog, etc. The data granularity consists of time series of annual, quarterly, monthly and other (weekly, daily and hourly) data. Data source: https://forecastingdata.org/ .

settings of their source works. All our models are implemented based on PyTorch, the open source code of each original paper is collected and experimented with, and all experiments are repeated 5 times and trained on dual NVIDIA TITAN X pascal 12-GB-GPU workstations. After referring to the research results in Section 5, MAE, RMSE, and MAPE were used as evaluation metrics.

7.1. Main results and analysis

After conducting experiments on the ETTh1 dataset, the obtained results are shown in Table 12, where the two best experimental results produced under each horizon of the dataset are bolded. On the ETTh1 dataset, the MTGNN and FDEformer method perform better when the prediction horizon is small, but as the prediction horizon grows, Autoformer gradually outperforms the MTGNN, which shows that on ETTh1, the FEDformer and Autoformer models are more suitable for LSTF.

After conducting experiments on the Stock dataset and COVID19 dataset, as shown in Table 13, we find that the mechanisms of progressive trending and seasonal decomposition yield good results in the face of multiple periodicities and trends when the dataset has few relevant features or repetitions. Autoformer and FEDformer achieve far better performance than the other models on the Stock dataset and the COVID19 dataset. Both datasets contain multiple repeated cycles and trends that are harder to capture. The Stock dataset contains cyclical changes in stock days, weeks, months, quarters, years, etc., and is also affected by unusual events such as the 2008 financial crisis. COVID19, on the other hand, has a low data volume, but it possesses the characteristic of an extremely strong exponential growth trend at the beginning that then exhibits volatility and a decreasing trend with seasonal and yearly changes. Both the Autoformer and FEDformer models apply the characteristics of asymptotic trending and seasonal decomposition, and they both exploit the characteristics of time series in the frequency domain, and we believe that this mechanism is the key to their far superior performance on the Stock and COVID19 datasets.

Experimenting on the PEMS03 dataset, as shown in Table 14, we find that the Transformer-type model is more suitable for LSTF than the

MTGNN-type model when many data feature variables and complex correlations are present. In the PEMS03 dataset, the spatial topology information is obvious, and we take the data of each node as a feature, so the dataset has 358 new features; thus, it has more features than the other datasets (ETTh1 has 7 feature variables, and Stock has 12 feature variables). The MTGNN spatio-temporal prediction model and the Transformer source model have better performance on this dataset, and the correlations between the data feature variables are complicated due to the presence of many data feature variables. While the MTGNN achieves better prediction results by adaptively learning the topological graph structure of the data, the Transformer yields better prediction results via fully self-attentive dot product operations. The remaining Transformer-class models inevitably lose some of the dependencies among the data in cases with complex correlations among such data feature variables due to the reduction in the self-attentive computational complexity. This aspect is not obvious in other datasets with few feature variables but is gradually revealed in data with many feature variables, such as PEMS03. As the prediction horizon grows, Reformer gradually outperforms the MTGNN, and it can be seen that the models with better LSTF performance on this dataset are Reformer and the Transformer, which are the two models with the highest computational time complexity levels among the selected Transformer-class models.

On the WTH dataset, we find that Reformer and Pyraformer have good LSTF performance. However, since there is also more obvious spatial topological information and spatial correlations in the WTH weather data, the MTGNN has good performance when the prediction horizon is short, but as on the other datasets, the characteristics of the MTGNN gradually lag behind those of the Transformer-type models as the prediction horizon increases.

In summary, we obtain the following conclusions from our experiments.

- (1) Spatio-temporal prediction models such as the MTGNN achieve great results on spatially correlated datasets such as traffic datasets and weather datasets, but their LSTF results on these datasets are not as good as those of Transformer-type models.

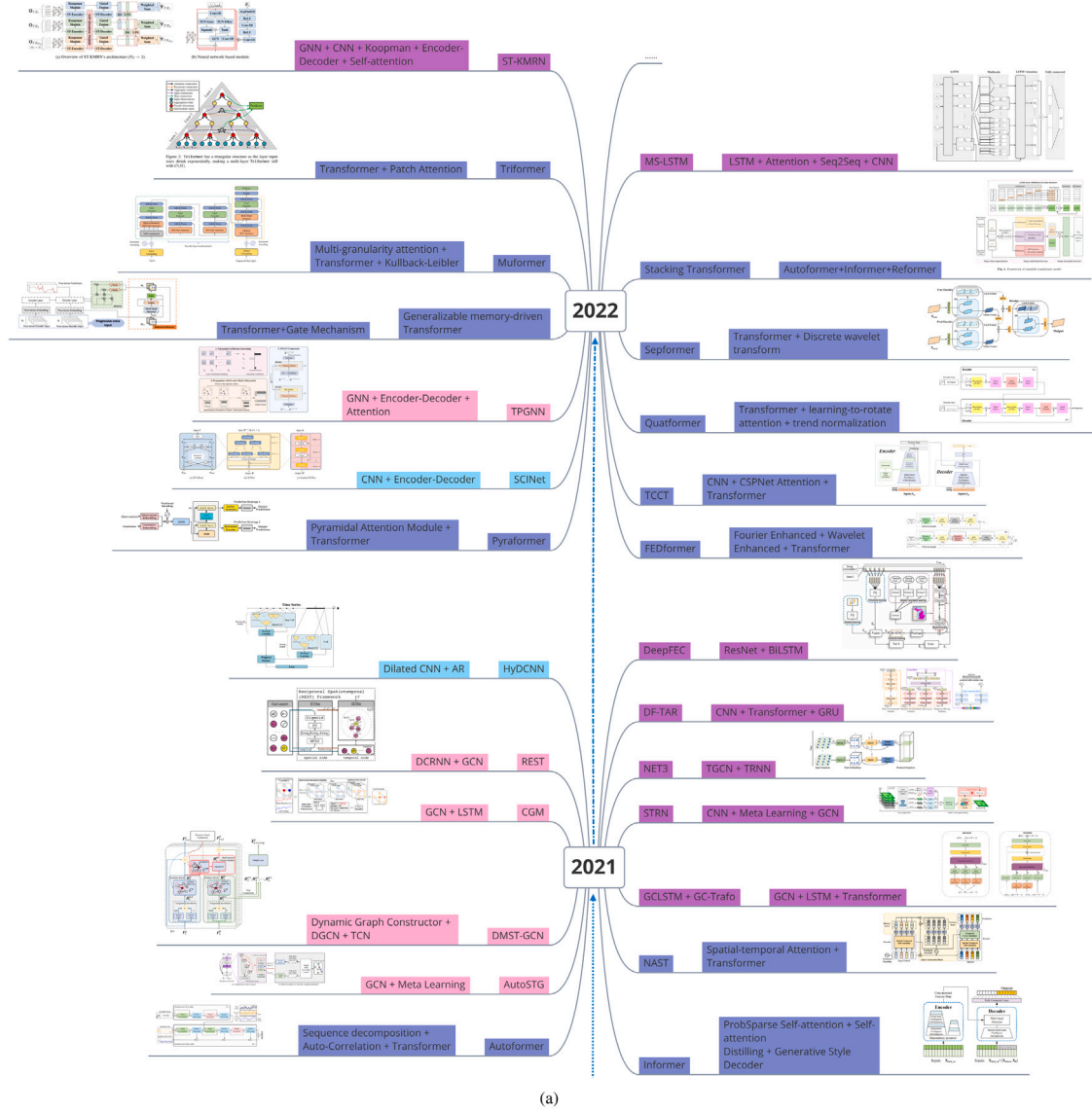


Fig. B.15. The Evolution of LSTF Networks.

- (2) The progressive trend and seasonal decomposition mechanism is more suitable for cases where few relevant features are contained in the dataset and there are multiple repeated cycles and trends among the data. The characteristics of time series in the frequency domain are worth studying.
- (3) Transformer-based models with many data feature variables and complex correlations inevitably lose some of the dependencies between data while reducing the computational complexity of self-attentiveness, thus leading to a decrease in prediction accuracy.
- (4) Transformer-based models are more practical for LSTF. According to the above experimental results, it can be seen that the Transformers model achieves better results on LSTF for most of the datasets.

7.2. Performance comparisons among long sequence time-series forecasting models

After the above experiments conducted with nine models on each of the five described datasets, how can we compare their LSTF performance? We use the theory proposed in Section 4.4 to illustrate the performance from two perspectives: model-model and model-self.

Furthermore, we believe that a single numerical magnitude comparison between models in terms of evaluation metrics is not correct, and we need to demonstrate statistically significant differences between the predictions of *model 1* and *model 2* for a given prediction scenario. Therefore, we use the Kruskal–Wallis test to compare whether the data distributions of the differences between the true and predicted values of the different models are significantly different.

We illustrate the results of Informer, Autoformer, Fedformer, Pyraformer, Reformer, Transformer, and LSTNet on the ETTh1 and Stock datasets as examples. When the models make predictions with the predict horizon = 768 on the ETTh1 dataset and the predict horizon = 480 on the Stock dataset, we perform a Kruskal–Wallis test by comparing the difference between the model's predicted values and the true values. We use a two-sided test with a significance level of 0.05 to test whether there is a significant difference in the data distribution of the predicted results between the two models. The results are shown in Table 15.

From the model-model perspective, in Table 15, it can be observed that, in the case of a prediction horizon = 768 in ETTh1, Autoformer exhibits significant statistical differences in its predictive performance compared to other models in this prediction task. This suggests that Autoformer outperforms other models in terms of long-term prediction

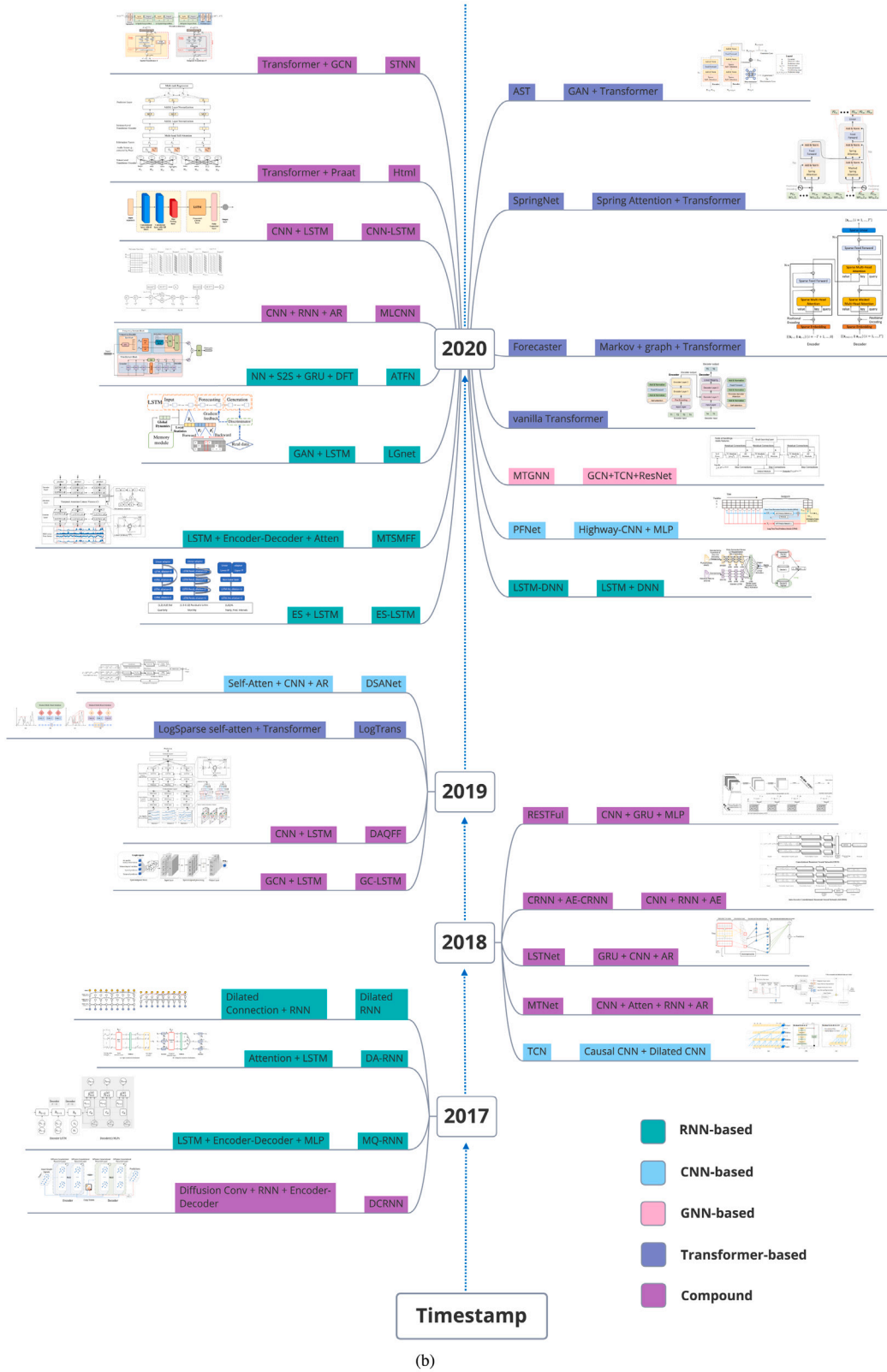


Fig. B.15. (continued).

performance in this task, while the significance level of the difference between the predictive results of Autoformer and FEDformer is greater than 0.05, indicating no significant difference between the two. Therefore, we conclude that the long-term predictive performance of

Autoformer and FEDformer is similar in this prediction task. Additionally, similar to the above, Autoformer significantly outperforms other models in the Stock dataset in this prediction task, with statistical differences, and its LSTF predictive performance is also similar to

that of FEDformer, with no significant differences observed in this prediction task.

8. Conclusion and future outlook

Based on our findings, we propose tentative future research directions for the field of LSTF as follows.

- (1) *At present, the compound model and Transformer-based model represent the main research directions in LSTF.* As evidenced by the findings of our research presented in Section 3, the majority of LSTF-related works in 2022 and 2021 have been built on the Transformer-based model and the Compound model, which are designed to capture the long-term dependencies present [18]. Currently, the Transformer-based model appears to be one of the most effective solutions in LSTF, as indicated by previous studies [17,33]. Additionally, the Compound model, which combines the strengths of various techniques to address the limitations of each, has also shown to be an effective approach in LSTF.
- (2) *Progressive trending and seasonal decomposition mechanisms should be introduced.* In real scenarios, cases often possess few relevant features in their datasets [174], and multiple cycles and trends are hidden and repeated among the data. We experimentally find that the progressive trend and seasonal decomposition mechanism [47] may be helpful in this case, which can be investigated in depth in the future.
- (3) *Currently, the research on Transformer-based models has some inherent limitations in LSTF.* The Transformer architecture is known for its high model capacity and ability to model long-term dependencies [111], making it a popular choice for LSTF. However, the Transformer-based models have some inherent limitations in LSTF, such as time complexity [91]. Most of the current optimization efforts for Transformers are also focused on addressing this issue [96,101]. However, our experiments have shown that Transformer-based models may unavoidably lose some of the dependencies between data points when reducing the complexity of self-attention computations in situations where there are many data feature variables with complex correlations, resulting in reduced prediction accuracy. Therefore, Transformer-based models are not suitable for all LSTF tasks, and other feature extraction frameworks can achieve performance that is comparable to Transformers [81]. Therefore, the choice of an appropriate feature extraction model should be based on the specific requirements of the task.
- (4) *Capturing dynamic graphs and long-term dependencies is critical for spatio-temporal long-term prediction.* In spatio-temporal long-term prediction, the hidden topological structure in the data typically undergoes significant changes as the prediction horizon increases [175]. Using static graphs for prediction can lead to significant biases [82], making the modeling of dynamic graphs essential for spatio-temporal long-term prediction. In addition, we have found that spatio-temporal prediction models that combine dynamic graphs, such as MTGNN, perform better on datasets with significant spatial correlations, such as transportation and weather, but their performance in long-term prediction is not as good as Transformer-based modules. This suggests that their model architecture is not optimal for feature extraction in long-term dependencies. Therefore, one of the future research directions for improving spatio-temporal long-term prediction is how to combine the characteristics of these two types of models.

In this article, we present a comprehensive overview of DL in LSTF. We provide deeply multi-angle definition explanations of LSTF and propose a new taxonomy of LSTF research based on the current mainstream feature extraction structures of models. We classify them into

RNN-based model, CNN-based model, GNN-based model, Transformer-based model, Compound model, and miscellaneous methods according to their main predictive network structure characteristics. To achieve a fair evaluation of LSTF performance, we apply the Kruskal–Wallis test and propose a new LSTF performance evaluation method from two perspectives. Additionally, we have collected abundant resources on TSF and LSTF, including datasets, application fields, evaluation metrics, and model comparison code libraries, which are provided in the open-source link. Finally, based on the results of our research and experiments, we summarize four possible future research directions.

Our review provides a comprehensive overview of the recent developments in LSTF and offers researchers valuable insights into improving their models with appropriate feature extraction methods. Our ‘Abundant Resources’ provide researchers with easy access to critical information on application domains, datasets, evaluation metrics, and LSTF method codes. Additionally, our newly proposed metrics can facilitate researchers in further analyzing and comparing the performance of LSTF models.

CRedit authorship contribution statement

Zonglei Chen: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Minbo Ma:** Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Tianrui Li:** Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Funding acquisition. **Hongjun Wang:** Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Funding acquisition. **Chongshou Li:** Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We give open source links to the data used in the article

Acknowledgments

This research was supported by the National Natural Science Foundation of China (Grant Nos. 62202395, 62176221, and 62276216), Natural Science Foundation of Sichuan Province (Grant No. 2022NS-FSC0930), and Fundamental Research Funds for the Central Universities, China (Grant No. 2682022CX067).

Appendix A. Datasets

The ‘-’ of time range means that the data has multiple different time ranges or that the time range of the dataset is maintained and dynamically updated.

A.1. Finance

See Table A.16.

A.2. Industrial energy

See Table A.17.

A.3. Urban transportation

See Table A.18.

A.4. Meteorological

See Table A.19.

A.5. Medicine

See Table A.20.

A.6. Competition

See Table A.21.

Appendix B. LSTF networks evolution

A finer-grained LSTF Networks Evolution is shown in Fig. B.15.

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